MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **DESIGN CALCULATION**
1800 SECURITY BASE CAMP
COMMUNITY BUILDING
DESIGN OF BUILDING STRUCTURE

Document Number **MD-216-1800-EG-CV-CAL-1547**
Revision: **D01**
Revision Date: **11.10.2023**
Status: **AS BUILT**

Outotec Document Number:	900707	GAU01	AN02	CED01	18002	OU100668731	2
L&T Document Number:	O19045	900	C	K	24248	001	2

REVISION HISTORY						
Rev	**Date**	**Revision Text**	**Prepared By**	**Reviewed By**	**Approved By**	**Employer**
A01	31.03.2020	FIRST ISSUE	MF	AVG	SJM	
C01	02.06.2020	ISSUED FOR CONSTRUCTION	MF	AVG	SJM	
D01	11.10.2023	AS BUILT	MF	AVG	SJM	

TABLE OF CONTENTS

DEFINITIONS AND ABBREVIATIONS 4
1. INTENT 5
2. EXECUTIVE SUMMARY 5
3. SCOPE 5
4. DESIGN METHODOLOGY & ASSUMPTIONS 6
4.1. Codes 6
4.2. Geotechnical investigations 7
4.3. Material Properties 7
5. DESIGN LOADS 7
5.1. Dead Load 7
5.1.1. Roof Dead Loads 7
5.1.2. Ground Floor Walls 9
5.2. Live Load 9
5.2.1. Roof Live Loads 9
5.3. Wind Load 9
5.4. Wind Load (Analytical Verification) 10
5.5. Seismic Load 15
5.6. Thermal/Temperature Loads (Self-restraining and Thermal Expansion and Contraction Loads) 15
6. DESIGN COMBINATIONS 16
7. CRACKED SECTIONS 17
8. DEFLECTIONS/DRIFT CHECK 18
8.1. Maximum Allowable Deflection 18
9. CONCRETE ROOF BEAM DESIGN 19
9.1. Design of Ledger Beam(LB1) 19
9.1.1. Staad Output For LB1 21
9.2. Design of Ledger Beam(LB2) 22
9.2.1. Long Term Deflection Check for LB2 Beam: 24

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

9.2.2. Staad Output For LB2 28
9.3. Design of Ledger Beam(LB3) 29
9.3.1. Staad Output For LB3 30
9.4. Design of Beam-(B1) 31
9.4.1. Staad Output For B1 32
10. CONCRETE GRADE SLAB DESIGN 33
11. DESIGN OF BEAM-(GB1) 34
11.1. DESIGN OF BEAM-(GB1) 34
11.1.1. STAAD OUTPUT FOR GB1 35
12. CONCRETE COLUMN DESIGN 36
12.1. Staad Output For Column C1 & C2 37
13. CANTILEVER SLAB DESIGN 38
14. FOUNDATION DESIGN 40
References 41
ATTACHMENT 1 STAAD INPUT DATA 42
ATTACHMENT 2 FOUNDATION DESIGN CALCULATIONS 48

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
Typ.	:	Typical
HCS	:	Hollow Core Slab
Fc'	:	Concrete Compressive Strength
Fy	:	Yield Strength of Reinforcing Steel
ϕ	:	Strength Reduction Factor
SBC	:	Saudi Building Code
SASO	:	Saudi Standards, Metrology and Quality Organization
ACI	:	American Concrete Institute
ASCE	:	American Society of Civil Engineers
PCI	:	Precast/Prestressed Concrete Institute

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

1. INTENT

This Document is intended to produce the technical needs in Engineering including particular reference to design justification & Engineering validation works. The work described below is related to Design of Community Building in Security Base Camp Area.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The Scope of this document is limited to Complete Analysis & Structural Design Verification of Community Building only.

PROPRIETARY & CONFIDENTIAL

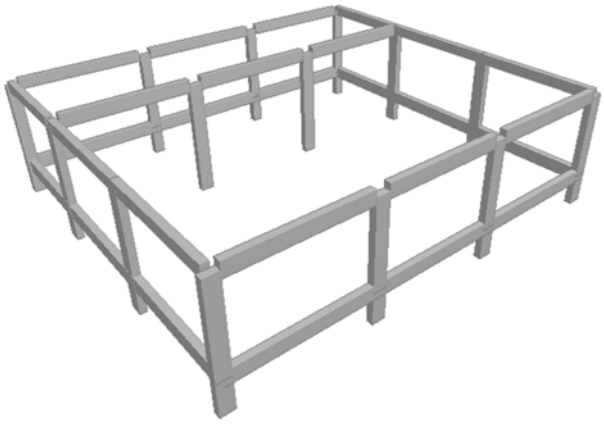
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

4. DESIGN METHODOLOGY & ASSUMPTIONS

This Building Is a Single Story Frame Structure with Foundation Being Cast in Site. The Building Shall Have Insulated Block Walls and Cast in Site Framing Beams to support the HCS (Hollow Core Slab)

The Superstructure for this Accommodation Building was modeled using STAAD software. The Analysis of Frames (Cast-In-Situ Columns & Beams) has been performed meeting design requirements & also keeping in view the Architectural requirements. The loads were applied as per SBC/ASCE/ACI standards & analysis was done to generate the Column Reactions at the Base which are assumed to be fixed. The Foundation design has been done by importing Column Base Reactions into Foundation Design Software.



STAAD 3D MODEL

4.1. Codes

[1] SBC-301-2007-Structural Loading & Forces.

[2] SBC-303-2007-Structural Soil & Foundations.

[3] SBC-304-2007-Structural Concrete Structures.

[4] SASO SSA 2 – Steel Bars for the Reinforcement of Concrete.

[5] SASO SSA 224 – Steel Fabric for Reinforcement of Concrete

[6] ASCE-7-2005 Minimum Design Loads for Buildings and other Structures.

[7] ACI 318-08 Building Code Requirements.

[8] PCI –MNL-120-04 –Design Handbook, 6th edition.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

4.2. Geotechnical investigations

Foundations have been designed based on recommendations of the Soil Investigation Report, Doc # MD-216-3000-EG-CV-RPT-1008.

4.3. Material Properties

Compressive Strength of Concrete at 28th days

fc' = 35 MPa (5.00 Ksi) for beams, columns, suspended slabs.

fc' = 35 MPa (5.00 Ksi) for foundations, grade beams, columns.

Yield Strength of Steel

fy = 414 MPa (60 Ksi) for all reinforcing steel

5. DESIGN LOADS

5.1. Dead Load

5.1.1. Roof Dead Loads

- Provision for Electrical Lighting = 0.05 kN/m²

Ceiling

Suspended steel channel system	= 0.10 kN/m²
Acoustical Fiber Tile (16mm)	= 0.05 kN/m²
Sub-total	= 0.2 kN/m²

- Hollow Core Slab (PCI 6th Edition)

(HCS-300 Topped) b/w Grid 2 & 4	= **3.88 KN/m2**
(HCS-150 Topped) b/w Grid 1 & 2	= **2.52 KN/m2**

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Prestressed Hollowcore Floor										
Overall Structural Depth (mm)	Self Wt- KN/M2	No. of Cores in Unit	Spans indicated below allow for Characteristic Service Load (Live Load) Plus Self Weight Plus 1.5Kn/m² for Finishes Characteristic Service Loads Kn/m²							
			0.75	1.50	2.00	2.50	3.00	4.00	5.00	10.00
			Clear Span in metres							
100	2.38	0	5.39	5.19	5.08	4.91	4.72	4.40	4.14	3.30
150	2.52	11	7.40	7.40	7.40	7.40	7.28	6.80	6.40	5.12
150 (Dense	2.94	11	7.40	7.40	7.40	7.35	7.08	6.64	6.27	5.06
200	3.10	11	9.40	9.40	9.40	9.40	9.26	8.69	8.21	6.65
250	3.43	6	10.80	10.80	10.80	10.80	10.80	10.80	10.56	8.60
300	3.88	6	13.50	13.50	13.50	13.50	13.18	12.44	11.81	9.70
350	4.23	6	15.00	15.00	15.00	15.00	14.89	14.08	13.39	11.06
400	4.57	6	16.00	16.00	16.00	16.00	15.97	15.13	14.42	11.98
450	4.92	6	18.00	18.00	18.00	17.70	17.19	16.33	15.58	13.02
500	5.27	6	18.00	18.00	18.00	18.00	18.00	17.13	16.38	13.76

- Built-in Water Proofing System

100mm thk Polystyrene Foam

0.01 kN/m² per 25.4mm thickness x (100/25.4mm) = 0.039 kN/m²

Water proofing membrane

Bituminous, smooth surface = 0.07 kN/m²

90 mm thick average cement foam concrete screed Lightweight

Concrete 0.38 kN/m² per 25mm thickness x (90/25mm) = 1.368 kN/m²

75 mm Concrete screed above HCS = 21 x 0.075 =1.6 kN/m²

Sub-total = **3.1 kN/m²**

Total Load b/w Grid 2 & 4 = **7.18 kN/m²**

Total Load b/w Grid 1 & 2 = **5.82 kN/m²**

UDL on External Beams along Grid Line-4 = 10.1 x 7.18/2 =**36.25 Kn/m**

UDL on Internal Beam along Grid Line-2 = 36.25+(4.25*5.82/2) =**48.62 Kn/m**

UDL on External Beams along Grid Line-1 = 4.25*5.82/2 =**12.36 Kn/m**

Moment on all the External Beams due to Cantilever Portion of slab and Screed

= (0.15x24+ (0.05+0.2)/2x20)*0.6*0.6/2 = **1 knm/m**

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

5.1.2. Ground Floor Walls

- 250 thk. Outer CMU

16 x 0.25 x 3.0m clear height = **12.0 kN/m**

5.2. Live Load

5.2.1. Roof Live Loads

Live loads were taken from SEI/ASCE 7-05, Table 4-1.

Ordinary Flat = **0.50 kN/m²**

UDL on External Beams along Grid Line-1 = 0.50 x 4.25/2 =**1.06 Kn/m**

UDL on External Beams along Grid Line-4 = 0.50 x 10/2 =**2.5 Kn/m**

UDL on Internal Beams along Grid Line-2 =**3.56 Kn/m**

Moment on all the External Beams due to Live Load on

Cantilever Portion of slab = 1*0.6*0.6/2 = **0.18 kN/m /m**

5.3. Wind Load

Wind loads were automatically generated in STAAD as per ASCE 7-05. The parameters used for the wind load generation are as follows (Refer STRUCTURAL DESIGN CRITERIA NON PROCESS BUILDINGS Doc No MD-216-0000-EG-ST-CRT-0001)

All structures and foundations (except for buried structures) designed and shall be constructed to resist the wind effects determined in accordance with Section-7.2, SBC-301-Analytical Procedure.

PROPRIETARY & CONFIDENTIAL

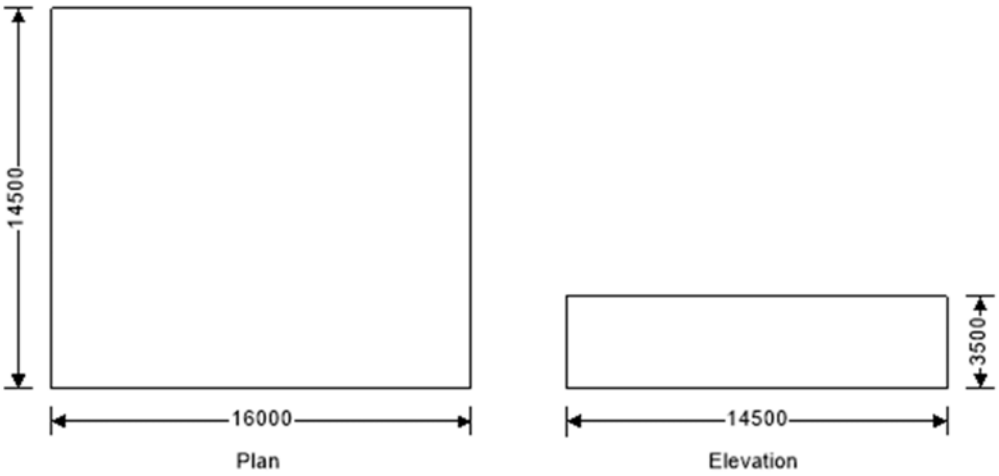
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

5.4. Wind Load (Analytical Verification)

WIND LOADING (SBC-301)

Using the directional design method



Building data

Type of roof;	Flat
Length of building;	b = 16000 mm
Width of building;	d = 14500mm
Height to eaves;	H = 3500 mm
Mean height;	h = 3500 mm

General wind load requirements

Basic wind speed;	V = **49** m/s (or) 177 kmph (Refer Clause-4.2.6, Doc- MD-216-1000-EG-ST- CRT-1001)
Risk category;	II(As per SBC-301, Table 1.6-1)
Velocity pressure exponent coeff;	K_d = 0.85(As per SBC-301, Table 6.4-1)
Exposure category ;	D(As per SBC-301, Table 6.4.2.3)
Enclosure classification ;	Partially enclosed buildings(As per SBC-301, Section-6.2)

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Internal pressure coef +ve; GC_{pi_p} = 0.55(As per SBC-301, Figure-7.2-1)

Internal pressure coef –ve; GC_{pi_n} = -0.55(As per SBC-301, Figure-7.2-1)

Gust effect factor; G_f = 0.85(As per SBC-301, Section-7.2.7.1)

Importance Factor; 1(As per SBC-301, Section-9.5)

Topography

Topography factor not significant; K_{zt} = 1.0(As per SBC-301, Section-6.4.3.2)

Velocity pressure equation; $q = 0.0473x10^{-3}Kn/m^2 \times K_z \times K_{zt} \times K_d \times V^2$ xI

V in Km/hr(As per SBC-301, Section-7.2.8)

Velocity pressures table

z (mm)	K_z (SBC-301,Table 7.2-2)	q_z (kN/m^2)
3500	1.03	1.29
3500	1.03	1.29

Pressures and forces

Net pressure; $p = q \times G_f \times C_{pe} - q_i \times GC_{pi}$;

Net force; $F_w = p \times A_{ref}$;

Walls load case 1 - Wind 0, GC_{pi} 0.55, - c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.50	0.80	1.29	0.17	55.99	9.38
B	3.50	-0.50	1.29	-1.26	56.00	-70.36
C	3.50	-0.70	1.29	-1.48	50.75	-74.88
D	3.50	-0.70	1.29	-1.48	50.75	-74.88

Walls load case 2 - Wind 0, GC_{pi} -0.55, + c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.50	0.80	1.29	1.58	55.99	88.73

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
B	3.50	-0.50	1.29	0.16	56.00	9.02
C	3.50	-0.70	1.29	-0.06	50.75	-2.94
D	3.50	-0.70	1.29	-0.06	50.75	-2.94

Walls load case 3 - Wind 90, GC_{pi} 0.55, - c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.50	0.80	1.29	0.17	50.74	8.50
B	3.50	-0.48	1.29	-1.23	50.75	-62.61
C	3.50	-0.70	1.29	-1.48	56.00	-82.62
D	3.50	-0.70	1.29	-1.48	56.00	-82.62

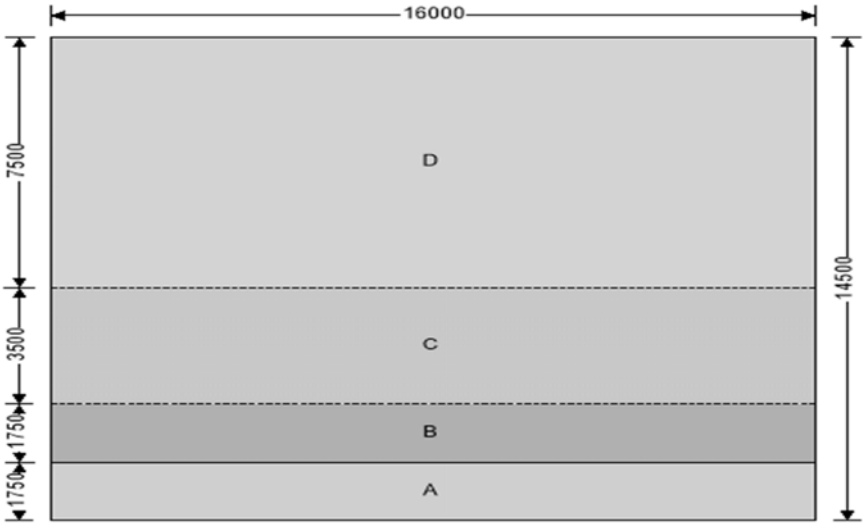
Walls load case 4 - Wind 90, GC_{pi} -0.55, + c_{pe}

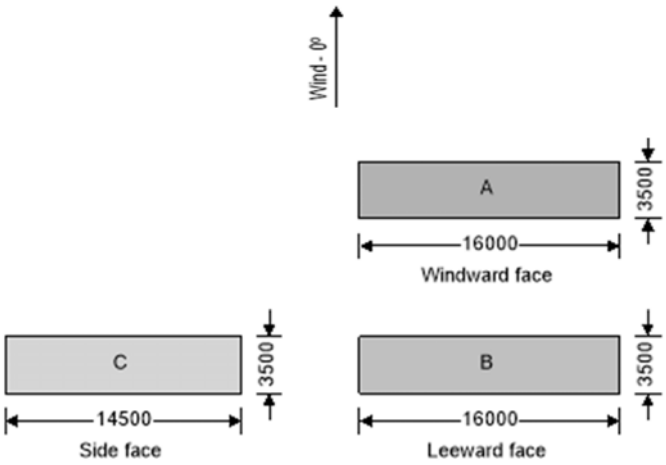
Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.50	0.80	1.29	1.58	50.74	80.42
B	3.50	-0.48	1.29	0.18	50.75	9.32
C	3.50	-0.70	1.29	-0.06	56.00	-3.25
D	3.50	-0.70	1.29	-0.06	56.00	-3.25

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

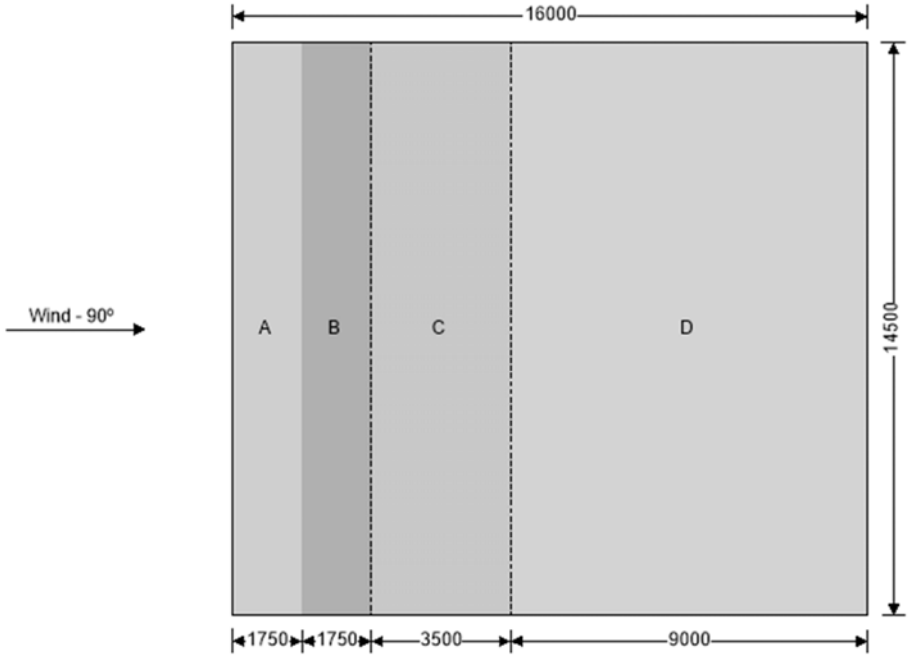




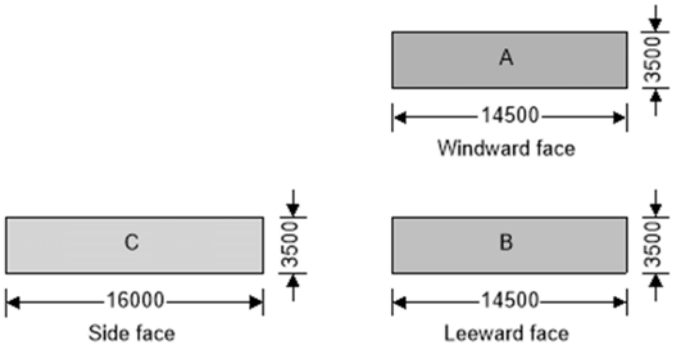
PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.



Plan view - Flat roof



So from the above analysis, the Wind Loads generated in STAAD are in accordance with SBC-301-2007-Structural Loading & Forces.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

5.5. Seismic Load

Earthquake loads were automatically generated in STAAD as per IBC 2003. The basic parameters affecting the generation of seismic loads is shown below:

Importance Factor, I = 1.0(As per SBC-301, Section-9.5)

Design Spectral Response Acceleration (Refer Clause-4.2.7b, MD-216-1000-EG-ST- CRT-1001)

Ss = 0.1g (Spectral Response Acceleration 0.2 second)

S1 = 0.02g (Spectral Response Acceleration 1 second)

S_{D_1} = 2/3*Fv*S1 = 2/3 * 2.4 * 0.02 = 0.032 (As per SBC-301, Section-9.4.4)

S_{DS} = 2/3*Fa*S1 = 2/3 * 1.60 * 0.10 = 0.10 (As per SBC-301, Section-9.4.4)

Where S_{D_1} Damped Design Spectral Response Acceleration@1 second

S_{DS} Damped Design Spectral Response Acceleration@short period

Site Class = D

Site Coefficient, Fa = 1.60 (As per SBC-301, Table 9.4.3a)

Site Coefficient, Fv = 2.40 (As per SBC-301, Table 9.4.3b)

Building period Coefficient, Ct=0.04 (As per SBC-301, Table 10.9.3.2)

For Ordinary reinforced concrete moment frame,

Response Modification Factor, Rx = 2.5 (As per SBC-301, Table 10.9.3.2)

Response Modification Factor, Rz = 2.5 (As per SBC-301, Table 10.2)

System Over Strength Factor, Omega = 3 (As per SBC-301, Table 10.2)

Deflection Amplification factor, Cd = 2.5 (As per SBC-301, Table 10.2)

5.6. Thermal/Temperature Loads (Self-restraining and Thermal Expansion and Contraction Loads)

An ambient temperature ranges from a maximum of 50° C to a low of 0° C. In direct exposure to sunlight, the temperature shall be assumed to rise a minimum of 15° C above ambient. As a minimum, the temperature gradient for thermal loads shall be +25° C and -15° C with the most severe condition governing the design.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

6. DESIGN COMBINATIONS

Service Loads (Allowable Stress Design) (SBC-301, Section 2.4.1)

D

D + T

D +Lr

D +0.75Lr

D + (W or 0.7E)

D + 0.75W + 0.75Lr

0.6D + W

0.6D + 0.7E

Factored loads (Strength Design) (SBC-301, Section 2.3.2)

1.4D

1.4D+1.4T

1.2(D + T) + 0.5Lr

1.2D + 1.6Lr + 0.8W

1.2D + 1.6W + 0.5Lr

1.2D + 1.0E

0.9D + 1.6W

0.9D + 1.0E

Where:

D = Dead Load

Lr = Live Load

W = Wind Load

E = Seismic Load

T = Temperature Load

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

7. CRACKED SECTIONS

In lieu of a more accurate analysis of the effects of cracking of concrete, provisions of SBC-304-Section 10.11 was used.

For Beams:	I = 0.35Ig
For Columns:	I = 0.70Ig
For Walls:	
Un-cracked	I = 0.70Ig
Cracked	I = 0.35Ig
For Slabs	I = 0.25Ig

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

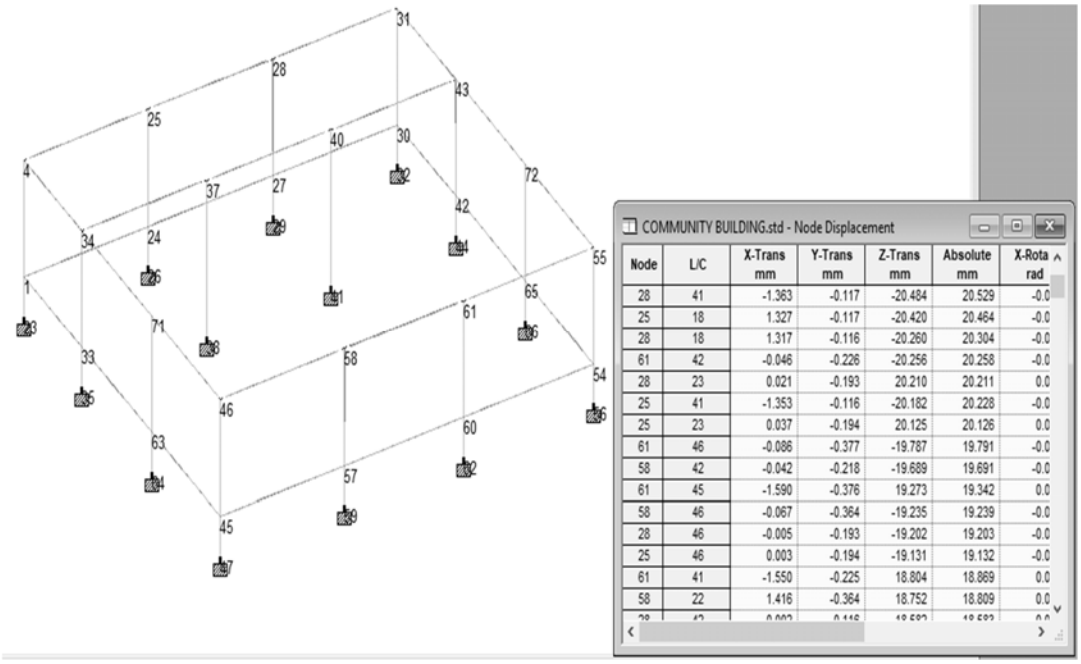
8. DEFLECTIONS/DRIFT CHECK

8.1. Maximum Allowable Deflection

L/240: for member load with live load plus dead load

L/600: for members supporting masonry under total loads

Where, L = span of member between support



Node	L/C	X-Trans mm	Y-Trans mm	Z-Trans mm	Absolute mm	X-Rota rad
28	41	-1.363	-0.117	-20.484	20.529	-0.0
25	18	1.327	-0.117	-20.420	20.464	-0.0
28	18	1.317	-0.116	-20.260	20.304	-0.0
61	42	-0.046	-0.226	-20.256	20.258	-0.0
28	23	0.021	-0.193	20.210	20.211	0.0
25	41	-1.353	-0.116	-20.182	20.228	-0.0
25	23	0.037	-0.194	20.125	20.126	0.0
61	46	-0.086	-0.377	-19.787	19.791	-0.0
58	42	-0.042	-0.218	-19.689	19.691	-0.0
61	45	-1.590	-0.376	19.273	19.342	0.0
58	46	-0.067	-0.364	-19.235	19.239	-0.0
28	46	-0.005	-0.193	-19.202	19.203	-0.0
25	46	0.003	-0.194	-19.131	19.132	-0.0
61	41	-1.550	-0.225	18.804	18.869	0.0
58	22	1.416	-0.364	18.752	18.809	0.0

Maximum attained Drift as per Analysis from above figure = 20 mm

Allowable Drift as per Table-10.12, SBC-301-2007 is H/100 = 3500/100 = 35mm

Hence the Structure is safe considering drift check…

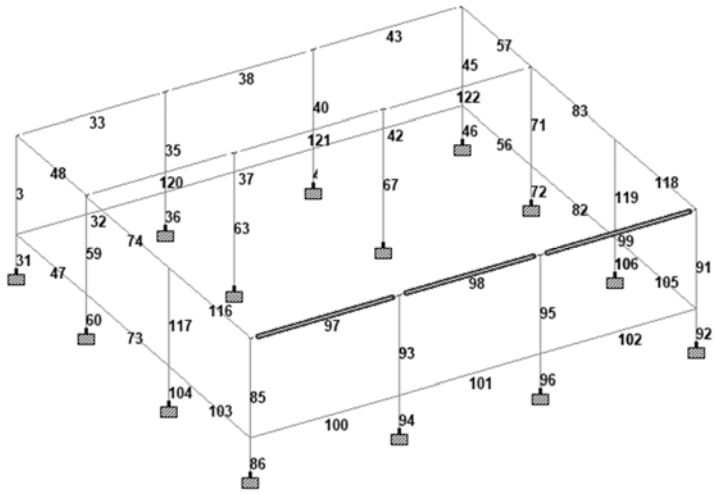
PROPRIETARY & CONFIDENTIAL

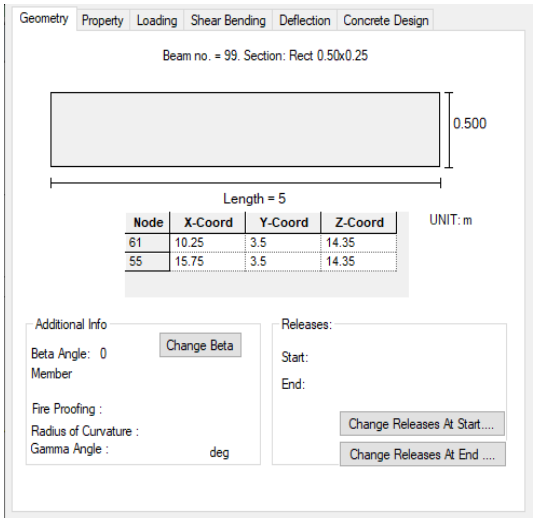
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

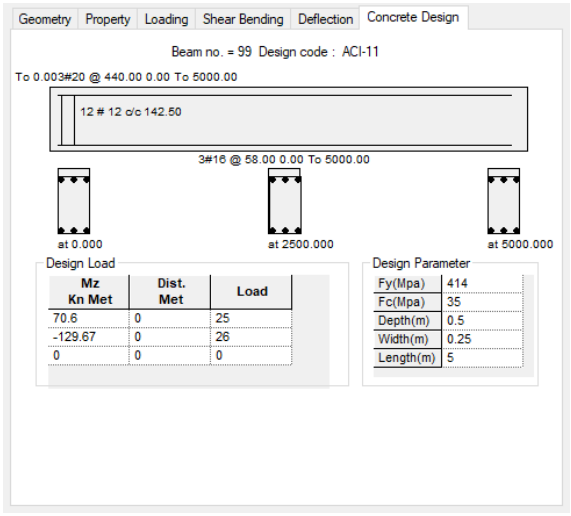
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

9. CONCRETE ROOF BEAM DESIGN

9.1. Design of Ledger Beam(LB1)







Design of Ledger Beam (LB1)

Maximum Moment required from Analysis = **129 Knm (Hogging)**

Therefore Ast.req =

$$A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$$

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Where b=250mm, Fy=414 Mpa, d=500-40-10-20/2=440mm, Fc'=35 Mpa & ϕ =0.9

Ast1.req = **830 mm²**

Ast.min=
(As per clause-10.5-ACI-318)

$$A_{s,min} = \frac{0.25\sqrt{f'_c}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

Ast.min= **471mm²**

Maximum Moment required from Analysis = **70 Knm (Sagging)**

Ast2.req = **440 mm²**

However Provide 3-20 dia -Full Straight Bars Top & 3-20 dia-Full Straight Bottom.

Ast1.pro**>>>>>>(Ast.min, Ast1.req)**

Ast2.pro**>>>>>>(Ast.min, Ast2.req)**

PROPRIETARY & CONFIDENTIAL

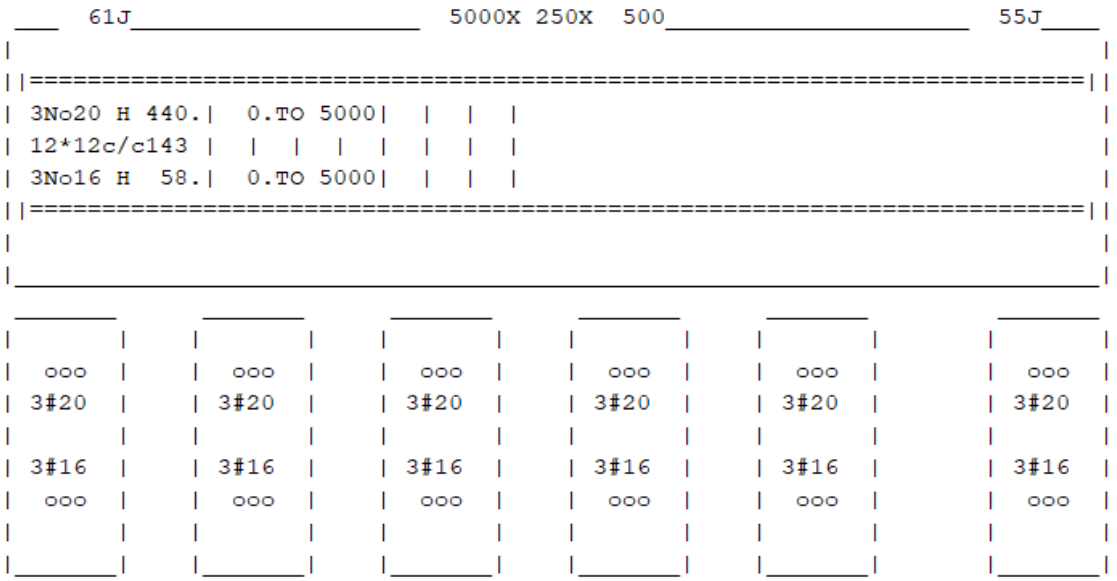
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

9.1.1. Staad Output For LB1

```
        B E A M  N O.    99 D E S I G N  R E S U L T S - SHEAR


   AT END   SUPPORT - Vu=   62.84 KNS  Tu=   24.6 KN-MET
                      Vc=  119.6 KNS, ACI 318:CLAUSE 11.6.3.1
 LOAD    49 TORSION VALUE TOO HIGH, INCREASE MEMBER SIZE.
   AT START SUPPORT - Vu=   60.70 KNS  Vc=  133.43 KNS  Vs=    0.00 KNS
   Tu=   22.57 KN-MET  Tc=    3.9 KN-MET  Ts=   30.1 KN-MET  LOAD    54
   STIRRUPS ARE REQUIRED FOR TORSION.
   REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
   PROVIDE 12 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 1560. MM
   ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   7.44 SQ.CM.
```

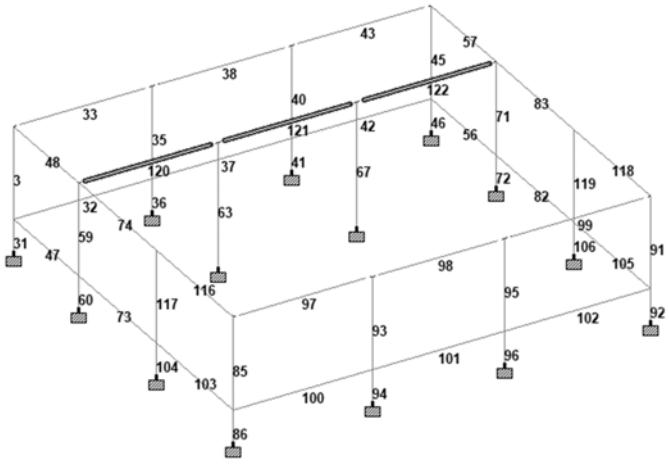


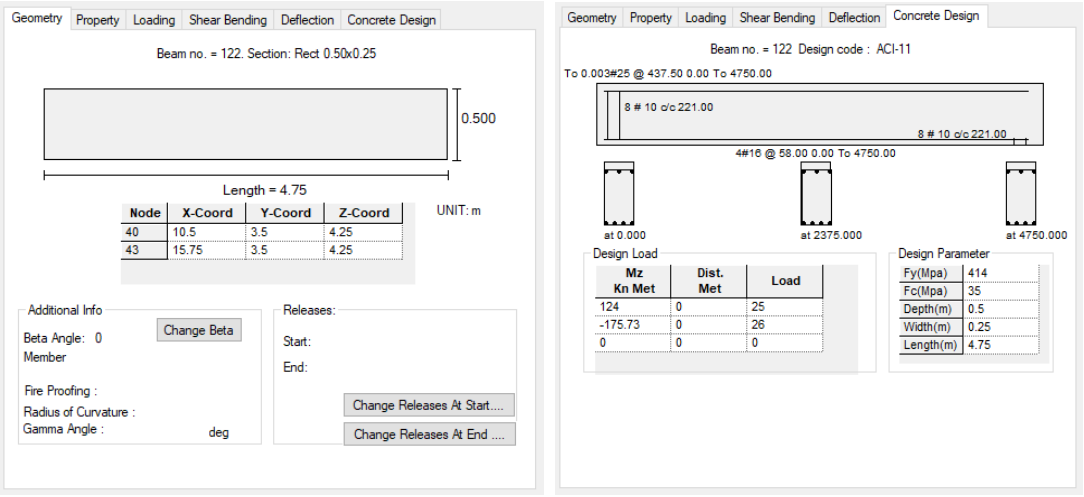
PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

9.2. Design of Ledger Beam(LB2)





Design of Ledger Beam (LB2)

Maximum Moment required from Analysis = **175 Knm (Hogging)**

Therefore $Ast._{req}$ =

$$A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$$

Where b=250mm, Fy=414 Mpa, d=500-40-10-20/2=440mm, Fc'=35 Mpa & ϕ =0.9

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Ast1.req = **1150 mm²**

Ast.min= (As per clause-10.5-ACI-318)

$$A_{s,\,min} = \frac{0.25\sqrt{f'_c}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

Ast.min= **471mm²**

Maximum Moment required from Analysis = **124 Knm (Sagging)**

Ast2.req = **794 mm²**

Hence Provide 3-25 dia-Full Straight Bars Top & 3-25 dia-Full Straight Bottom considering Long term deflection check as shown below in section 9.2.1.

Ast1.pro**>>>>>>(Ast.min, Ast1.req)**

Ast2.pro**>>>>>>(Ast.min, Ast2.req)**

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

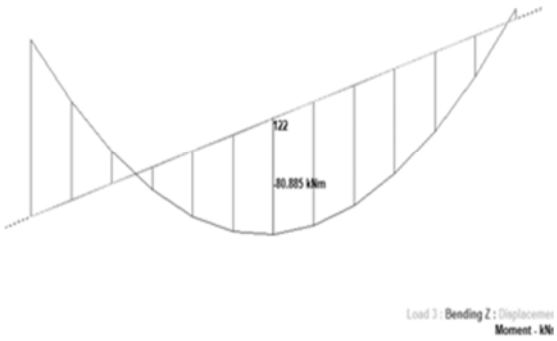
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

9.2.1. Long Term Deflection Check for LB2 Beam:

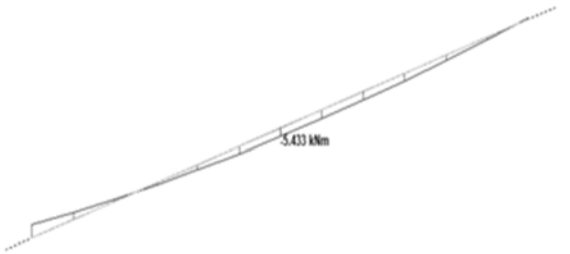
LONG TERM DEFLECTION CHECK:LB2

INPUT DATA :

Conc. Compressive stress, fc`	=	35.00	MPa
Reinforcement bar tensile stress, fy	=	414.00	MPa
Modulus Elasticity of concrete,	=	27,806	Mpa
Modulus Elasticity of steel, *Es*	=	200,000	Mpa
Unit wt. of concrete, Wconc	=	24.00	kN/m
Unit wt. of steel	=	77.00	kN/m
Depth of beam, h	=	500	mm
width of beam, b	=	250	mm
Length of beam, L	=	5.00	m
Bottom bar diam., db	=	25.00	mm
area of 1 bar Ab	=	490.87	mm^2
no. of bottom bars	=	3.00	pcs.
Top bar diam., db	=	25.00	mm
area of 1 bar Ab	=	490.87	mm
no. of top bars	=	3.00	pcs.
Stirrups diam. Dv	=	10.00	mm
area of 1 bar Av	=	78.54	mm
concrete cover, cc	=	40.00	mm
$d_{eff.bot}$ (@ bottom), d	=	437.50	mm
(FROM STAAD RESULTS) Mdead	=	80.00	kN-m
Mlive	=	6.00	kN-m



Load 3 : Bending Z : Displacemer
Moment - kN



Load 4 : Bending Z : Displacemer
Moment - kN

CHECK DEFLECTION: $M_{d+\lambda}$ = 86.00 kN-m

$$M_{sus} = M_d + 0.50 M_\lambda = 83.00 \text{ kN-m}$$

modulus of rupture, modulus of elasticity, modular ratio

$$f_r = 0.62\lambda\sqrt{f_c'} = 3.67 \text{ Mpa} \quad \leftarrow ACI(318M\text{-}08)\ Sec.\ 9.5.2.3$$

$$n_s = \frac{Es}{Ec} = 7.19$$

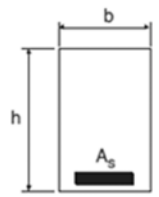
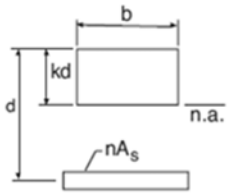
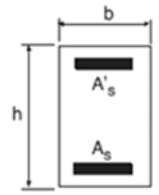
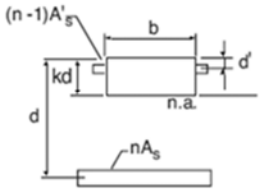
PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

gross and crack section of moment of inertia

Table 10-2 Gross and Cracked Moment of Inertia of Rectangular and Flanged Section

Gross Section	Cracked Transformed Section	Gross and Cracked Moment of Inertia
	 Without compression steel	$n = \frac{E_s}{E_c}$ $B = \frac{b}{(nA_s)}$ $I_g = \frac{bh^3}{12}$ Without compression steel $kd = \left(\sqrt{2dB+1}-1\right)/B$ $I_{cr} = b(kd)^3/3 + nA_s(d-kd)^2$
	 With compression steel	With compression steel $r = (n-1)A'_s/(nA_s)$ $kd = \left[\sqrt{2dB(1+rd'/d)+(1+r)^2}-(1+r)\right]/B$ $I_{cr} = b(kd)^3/3 + nA_s(d-kd)^2 + (n-1)A'_s(kd-d')^2$

Page 2

$$I_g = \frac{bh^3}{12} = 2.60E+09 \text{ mm}^4$$

$$B = \frac{b}{nAs} = 0.02 \text{ mm/mm}^2$$

$$kd = \left(\sqrt{2dB+1}-1\right)/B = 154.78 \text{ mm}$$

$$I_{cr} = \frac{b(kd)^3}{3} + nAs(d-kd)^2 = 1.16E+09 \text{ mm}^4$$

$$\frac{I_g}{I_{cr}} = 2.25$$

effective moment of inertia

$$M_{cr} = \frac{fr I_g}{y_t} = 38.21 \text{ kN-m} \quad \leftarrow ACI(318M\text{-}08)\ Sec.\ 9.5.2.3$$

under dead load

$$\frac{M_{cr}}{M_d} = 0.48$$

$$(I_e)_d = \left(\frac{M_{cr}}{M_d}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_d}\right)^3\right] I_{cr} = 1.31E+09 \text{ mm}^4 \quad \leftarrow ACI(318M\text{-}08)\ Sec.\ 9.5.2.3$$

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

under sustained load

$$\frac{M_{cr}}{M_{sus}} = 0.46$$

$$(I_e)_{sus} = \left(\frac{M_{cr}}{M_{sus}}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_{sus}}\right)^3\right] I_{cr} = 1.30E+09 \text{ mm}^4$$ ←ACI(318M-08) Sec. 9.5.2.3

under dead + live load

$$\frac{M_{cr}}{M_{d+\lambda}} = 0.44$$

$$(I_e)_{d+\lambda} = \left(\frac{M_{cr}}{M_{d+\lambda}}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_{d+\lambda}}\right)^3\right] I_{cr} = 1.28E+09 \text{ mm}^4$$ ←ACI(318M-08) Sec. 9.5.2.3

initial or short-term deflection

$$K = 1.00$$

$$(\Delta_i)_d = \frac{K(5/48)M_d\lambda^2}{Ec(I_e)_d} = 5.704 \text{ mm}$$

$$(\Delta_i)_{sus} = \frac{K(5/48)M_{sus}\lambda^2}{Ec(I_e)_{sus}} = 5.994 \text{ mm}$$

$$(\Delta_i)_{d+\lambda} = \frac{K(5/48)M_{d+\lambda}\lambda^2}{Ec(I_e)_{d+\lambda}} = 6.279 \text{ mm}$$

$$(\Delta_i)_\lambda = (\Delta_i)_{d+\lambda} - (\Delta_i)_d = 0.575 \text{ mm}$$

Table 10-3 Deflection Coefficient K

		K
1.	Cantilevers (deflection due to rotation at supports not included)	2.40
2.	Simple beams	1.0
3.	Continuous beams	1.2-0.2 M_o/M_a
4.	Fixed-hinged beams (midspan deflection)	0.80
5.	Fixed-hinged beams (maximum deflection using maximum moment)	0.74
6.	Fixed-fixed beams	0.60

For other types of loading, K values are given in Ref. 10.2.

M_o = Simple span moment at midspan $\left(\frac{w\ell^2}{8}\right)$.

M_a = Net midspan moment.

TABLE 9.5(b) — MAXIMUM PERMISSIBLE COMPUTED DEFLECTIONS

Type of member	Deflection to be considered	Deflection limitation
Flat roofs not supporting or attached to nonstructural elements likely to be damaged by large deflections	Immediate deflection due to live load *L*	$\ell/180^*$
Floors not supporting or attached to nonstructural elements likely to be damaged by large deflections	Immediate deflection due to live load *L*	$\ell/360$
Roof or floor construction supporting or attached to nonstructural elements likely to be damaged by large deflections	That part of the total deflection occurring after attachment of nonstructural elements (sum of the long-term deflection due to all sustained loads and the immediate deflection due to any additional live load)†	$\ell/480^‡$
Roof or floor construction supporting or attached to nonstructural elements not likely to be damaged by large deflections		$\ell/240^§$

*Limit not intended to safeguard against ponding. Ponding should be checked by suitable calculations of deflection, including added deflections due to ponded water, and considering long-term effects of all sustained loads, camber, construction tolerances, and reliability of provisions for drainage.

†Long-term deflection shall be determined in accordance with 9.5.2.5 or 9.5.4.3, but may be reduced by amount of deflection calculated to occur before attachment of nonstructural elements. This amount shall be determined on basis of accepted engineering data relating to time-deflection characteristics of members similar to those being considered.

‡Limit may be exceeded if adequate measures are taken to prevent damage to supported or attached elements.

§Limit shall not be greater than tolerance provided for nonstructural elements. Limit may be exceeded if camber is provided so that total deflection minus camber does not exceed limit.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

$$(\Delta_i)_\lambda \le \frac{\lambda}{180} = 27.78 \text{ mm} > 0.57 \text{ mm} \quad ok!$$

$$(\Delta_i)_\lambda \le \frac{\lambda}{360} = 13.89 \text{ mm} > 0.57 \text{ mm} \quad ok!$$

additional long-term deflection

$\xi = 2.00$

5 years or more 2.0
12 months.....1.4
6 months.....1.2
3 months.....1.0

$\rho = \frac{As}{bd} = 0.0135$

$\lambda_\Delta = \frac{\xi}{1+50\rho'} = 1.20$ ←ACI(318M-08) Sec. 9.5.2.5

Page 4

$\Delta_{cp} + \Delta_{sh} = \lambda(\Delta_i)_{sus} = 7.16$ mm

$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\lambda = 7.74$ mm

$$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\lambda \le \frac{\lambda}{480} = 10.42 \text{ mm} > 7.74 \text{ mm} \quad ok!$$

$$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\lambda \le \frac{\lambda}{240} = 20.83 \text{ mm} > 7.74 \text{ mm} \quad ok!$$

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

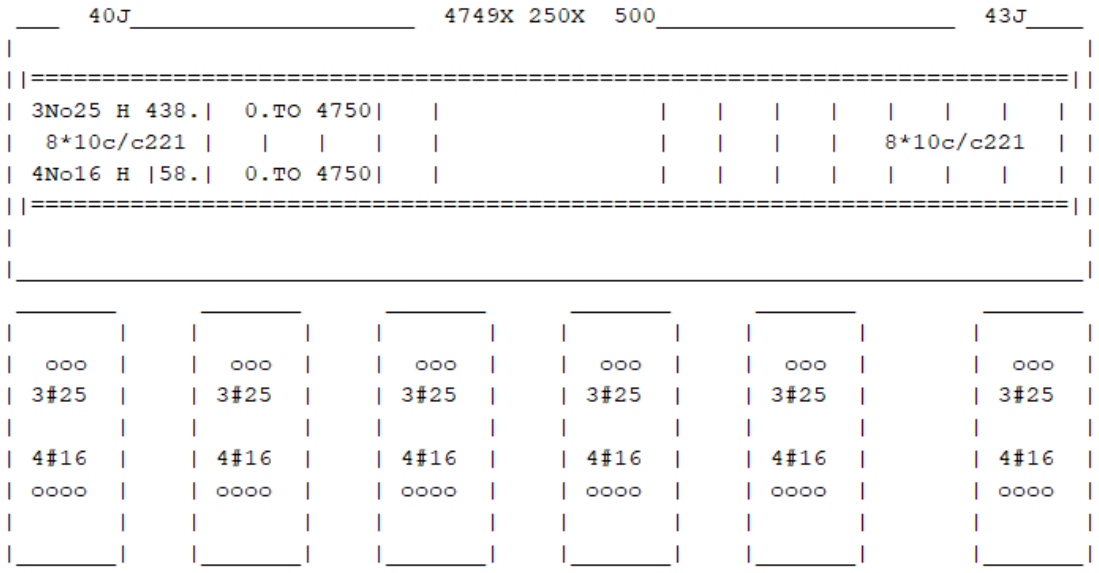
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

9.2.2. Staad Output For LB2

```
B E A M  N O.   122 D E S I G N  R E S U L T S - SHEAR


AT START SUPPORT - Vu=  121.42 KNS  Vc=  102.57 KNS  Vs=   59.32 KNS
Tu=    0.78 KN-MET  Tc=    3.6 KN-MET  Ts=    0.0 KN-MET  LOAD    28
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT IS REQUIRED FOR SHEAR.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM  C/C FOR 1435. MM

AT END   SUPPORT - Vu=   49.59 KNS  Vc=  108.11 KNS  Vs=    0.00 KNS
Tu=    0.04 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD     3
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM  C/C FOR 1435. MM
```

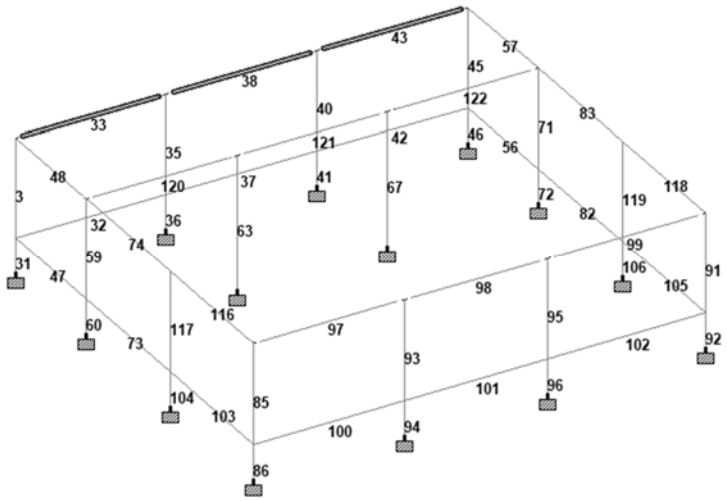


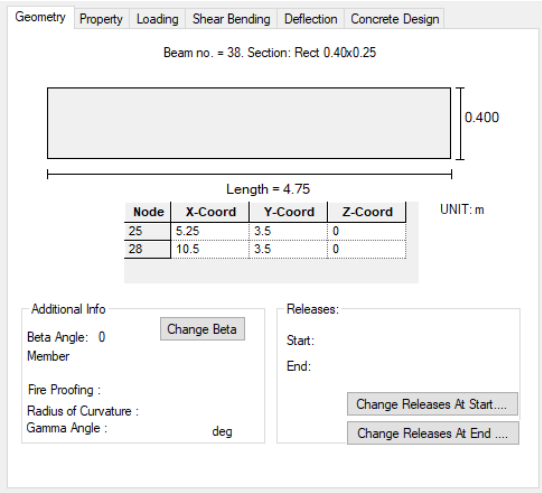
PROPRIETARY & CONFIDENTIAL

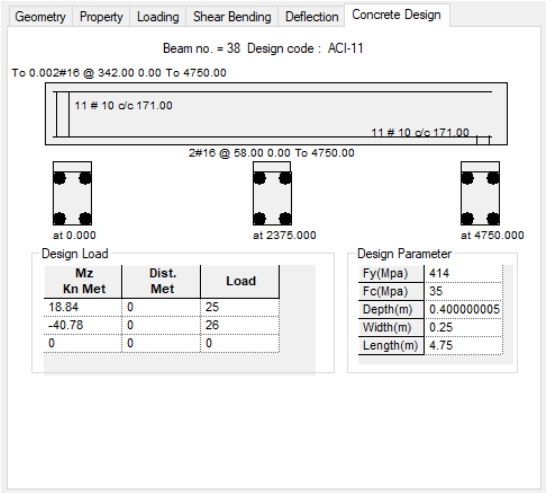
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

9.3. Design of Ledger Beam(LB3)







Hence Provide 3-16 dia-Full Straight Bars Top & Bottom considering minimum reinfrocement requirements

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

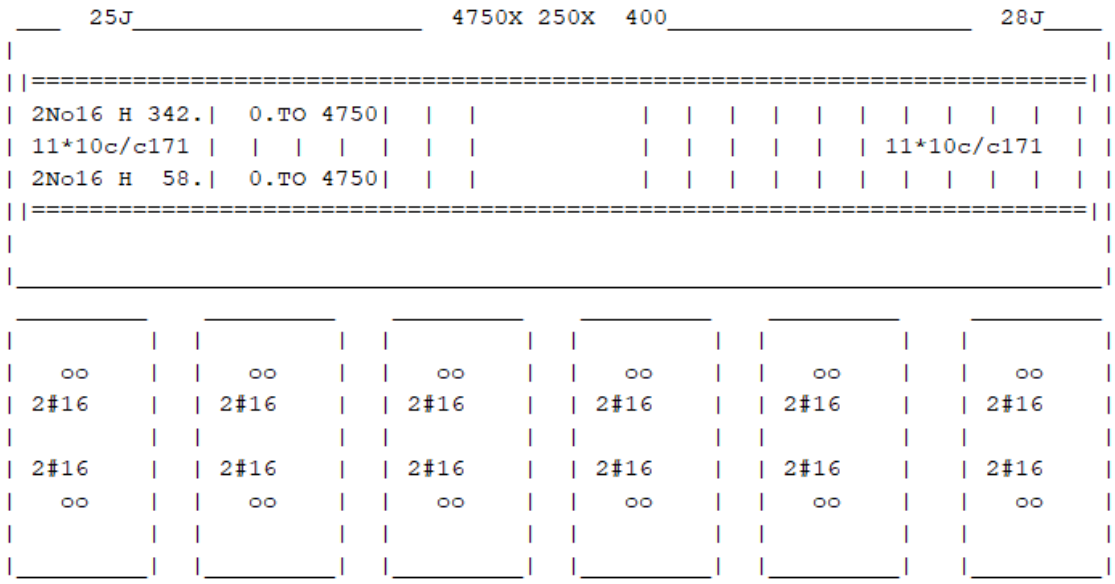
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

9.3.1. Staad Output For LB3

```
        B E A M  N O.    38 D E S I G N  R E S U L T S - SHEAR


AT START SUPPORT - Vu=   33.70 KNS  Vc=   85.40 KNS  Vs=    0.00 KNS
Tu=    2.31 KN-MET  Tc=    2.9 KN-MET  Ts=    0.0 KN-MET  LOAD    24
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 171. MM  C/C FOR 1635. MM

AT END   SUPPORT - Vu=   33.70 KNS  Vc=   85.40 KNS  Vs=    0.00 KNS
Tu=    2.27 KN-MET  Tc=    2.9 KN-MET  Ts=    0.0 KN-MET  LOAD    24
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 171. MM  C/C FOR 1635. MM
```

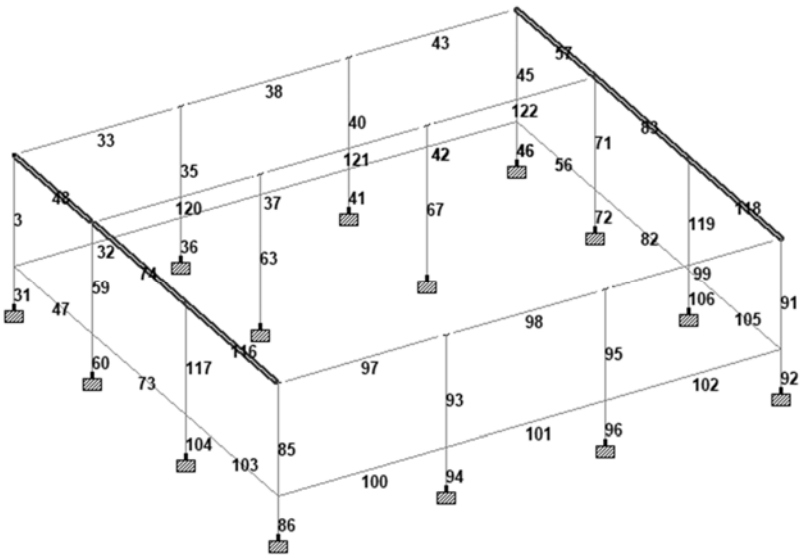


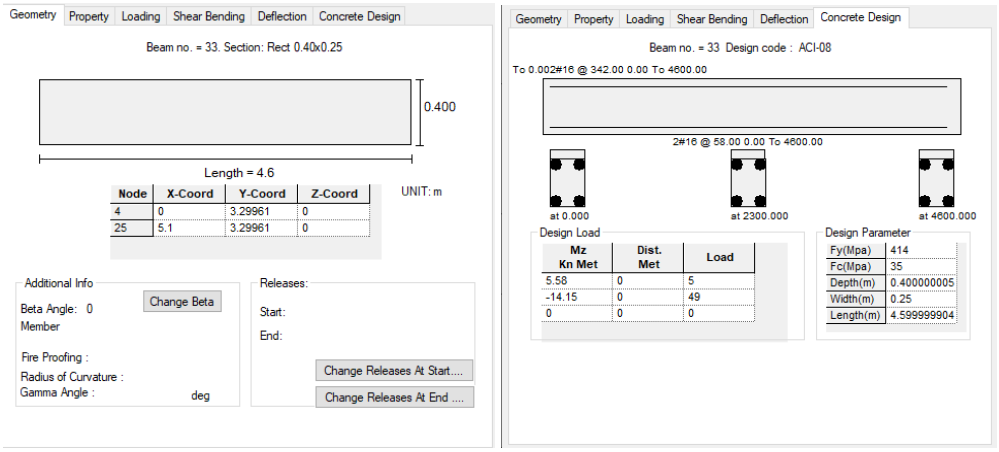
PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

9.4. Design of Beam-(B1)





Design of Beam (B1)

However Provide minimum 3-16 dia-Full Straight Bars Top & Bottom.

PROPRIETARY & CONFIDENTIAL

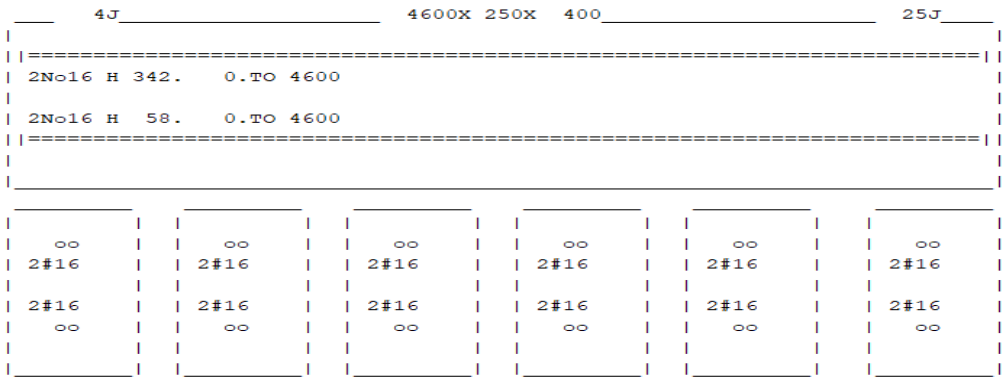
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

9.4.1. Staad Output For B1

```
            B E A M   N O.     33 D E S I G N   R E S U L T S - SHEAR


   AT START SUPPORT - Vu=     3.48 KNS   Vc=    82.47 KNS   Vs=     0.00 KNS
   Tu=     1.52 KN-MET   Tc=     2.9 KN-MET   Ts=     0.0 KN-MET   LOAD      3
                      STIRRUPS ARE NOT REQUIRED.
   AT END   SUPPORT - Vu=     3.42 KNS   Vc=    83.06 KNS   Vs=     0.00 KNS
   Tu=     1.40 KN-MET   Tc=     2.9 KN-MET   Ts=     0.0 KN-MET   LOAD      3
                      STIRRUPS ARE NOT REQUIRED.
```



PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

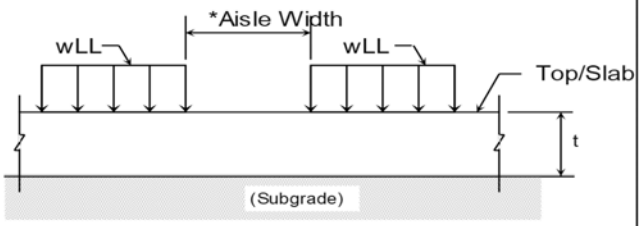
10. CONCRETE GRADE SLAB DESIGN

CONCRETE SLAB ON GRADE ANALYSIS

For Slab Subjected to Stationary Uniformly Distributed Live Loads

Input Data:

Slab Thickness, t =	3.937	in.	100.000	mm		
Concrete Strength, f 'c =	5000	psi	35.000	kpa		
Subgrade Modulus, k =	39.7866	pci	10800	Kn/m3	(AS PER SECTION-7.7-SOIL REPORT)	
Factor of Safety, FS =	1.500		1.500			
Uniform Live Load, wLL =	104.45	psf	5.00	kpa		



Concrete Slab on Grade with Uniform Loads

Results:

Design Parameters:

Modulus of Rupture, MR =	636.40	psi	4387.79	kpa	MR = 9*SQRT(f 'c)
Low. Bending Stress, Fb =	424.26	psi	2925.19	kpa	Fb = MR/FS
Modulus of Elasticity, Ec =	4030509	psi	27789329	kpa	Ec = 57000*SQRT(f 'c)
Poisson's Ratio, μ =	0.15		0.15		μ = 0.15 (assumed for concrete)
Radius of Stiffness, Lr =	26.94	in.	684.37	mm	Lr = (Ec*t^3/(12*(1-μ^2)*k))^0.25
Critical Aisle Width, Wcr =	4.96	ft.	1511.77	mm	Wcr = (2.209*Lr)/12

Stationary Uniformly Distributed Live Loads:

wLL(allow) =	682.05	psf	wLL(allow) = 257.876*Fb*SQRT(k*t/Ec)
	32.65	kpa	**wLL(allow) >= wLL, O.K.**

Provide Minimum Reinforcement of 0.14%@both faces = 0.14%(h) = 140 mm2/M

Table 24.4.3.2—Minimum ratios of deformed shrinkage and temperature reinforcement area to gross concrete area

Reinforcement type	f_y, psi	Minimum reinforcement ratio	
Deformed bars	< 60,000	0.0020	
Deformed bars or welded wire reinforcement	≥ 60,000	Greater of:	$\frac{0.0018 \times 60,000}{f_y}$
			0.0014

Provide WWF-8mm Dia@200 c-c Both Ways-TOP LAYER

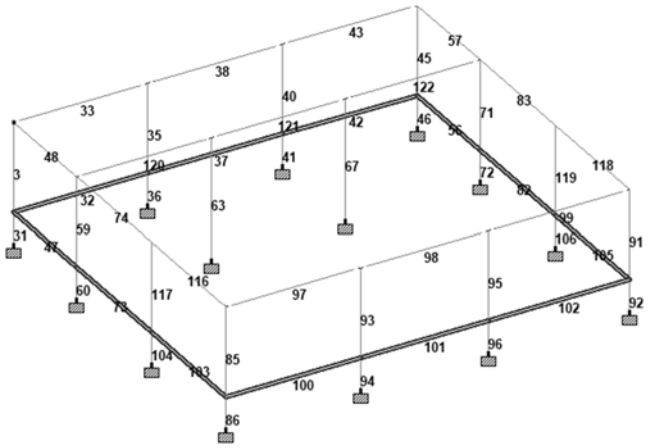
PROPRIETARY & CONFIDENTIAL

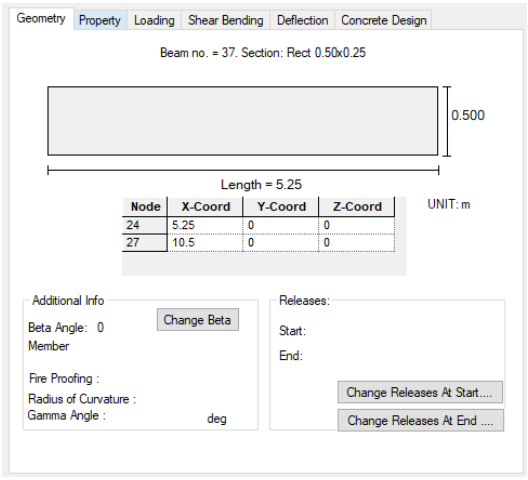
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

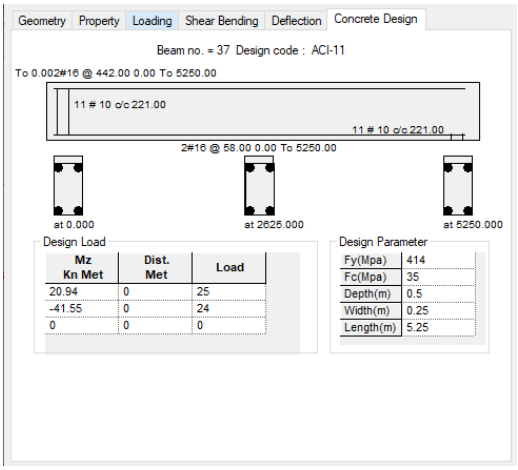
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

11. DESIGN OF BEAM-(GB1)

11.1. DESIGN OF BEAM-(GB1)







Provide minimum 3-16 dia-Full Straight Bars Top & Bottom.

PROPRIETARY & CONFIDENTIAL

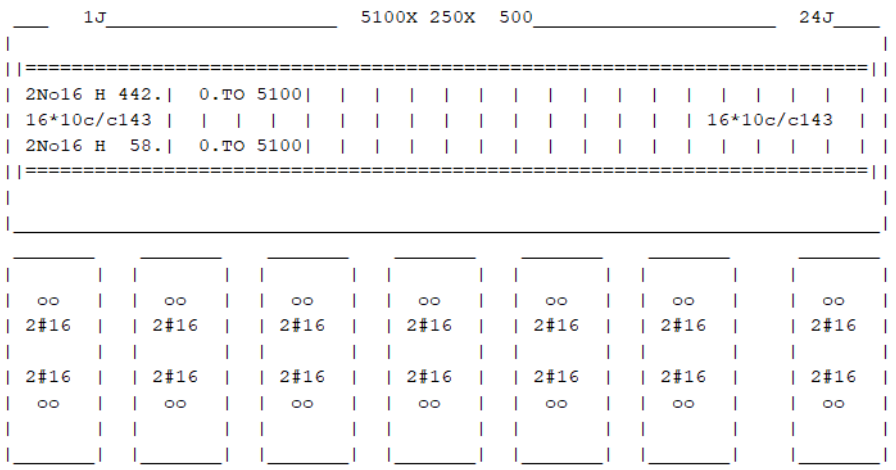
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

11.1.1. STAAD OUTPUT FOR GB1

```
AT START SUPPORT - Vu=   30.30 KNS  Vc=  105.94 KNS  Vs=    0.00 KNS
Tu=   14.42 KN-MET  Tc=    3.9 KN-MET  Ts=   19.2 KN-MET  LOAD    48
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 2110. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   4.75 SQ.CM.

AT END   SUPPORT - Vu=   35.28 KNS  Vc=  104.92 KNS  Vs=    0.00 KNS
Tu=   14.42 KN-MET  Tc=    3.9 KN-MET  Ts=   19.2 KN-MET  LOAD    48
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 2110. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   4.75 SQ.CM.
```

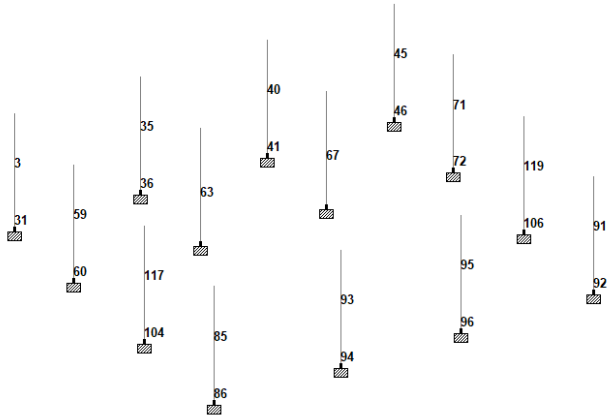


PROPRIETARY & CONFIDENTIAL

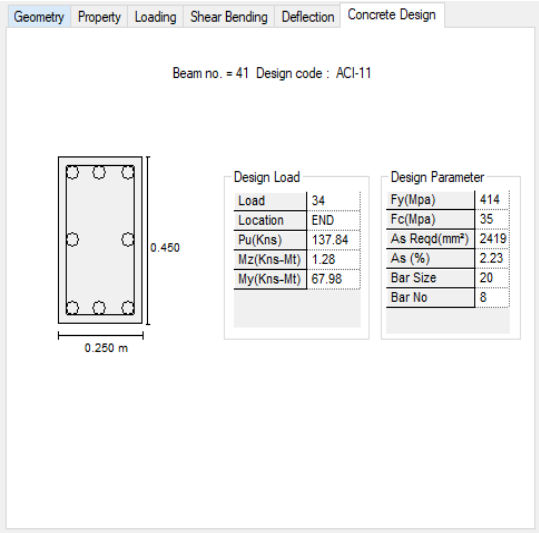
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

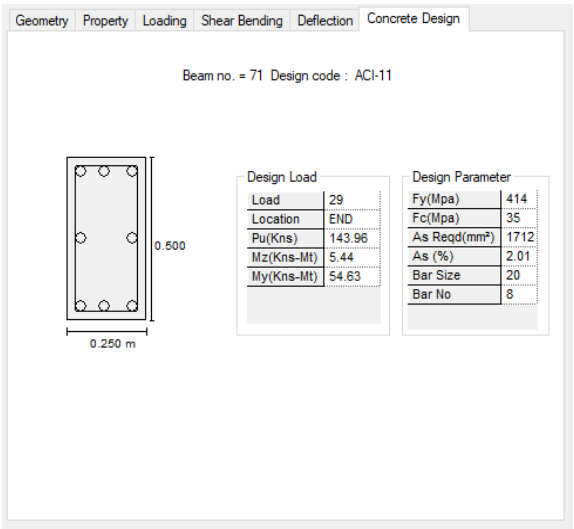
12. CONCRETE COLUMN DESIGN



Column numbers



Column Design C1



Column Design C2

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

12.1. Staad Output For Column C1 & C2

```
      ACI 318-11   COLUMN  NO.    41   DESIGN RESULTS
      ================================================


  FY - 413.7  FC -  35.0 MPA,  RECT SIZE - 250.0 X 450.0 MMS, TIED
         AREA OF STEEL REQUIRED =  2418.8  SQ. MM
-------------------------------< PAGE 38 Ends Here >--------------------------
   STAAD SPACE                                                -- PAGE NO.   39


  BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
  ----------------------------------------------------------

    8 - 20 MM              2.234       34      END     0.650
  (PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
  TIE BAR NUMBER   10 SPACING 250.00 MM


     ACI 318-11   COLUMN  NO.    71   DESIGN RESULTS
     ================================================


FY - 413.7  FC -  35.0 MPA,  RECT SIZE - 250.0 X 500.0 MMS, TIED
        AREA OF STEEL REQUIRED =  1712.5  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
----------------------------------------------------------

  8 - 20 MM              2.011       29      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 250.00 MM
```


PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

13. CANTILEVER SLAB DESIGN

DESIGN OF CANTILEVER SLAB

Geometrical Properties			Material Properties		
Span to center of support "sp_1"	0.8	met	Weight of reinforced concrete "γ_r	24	kN/m^3
Clear Span	0.6	met	Strength of concrete "f'_c"	35	N/mm^2
Clear span + depth of slab "sp_2"	0.75	met	Strength of steel "f_y"	420	N/mm^2
Max. eff span le = min (sp_1;sp_2)	0.8	met	Clear cover to concrete("c")	20	mm
Width of Slab "b"	1000	mm	Flexure strength reduction factor	0.9	
Thickness of slab "t"	150	mm	Shear strength reduction factor	0.75	
Eff. Depth "d"	125		Modulus of Elasticity of steel "Es"	2E+05	N/mm^2
Loading			Modulus of Elasticity of concrete	27806	N/mm^2
Self Weight of slab	3.6	kN/m^2	Modular Ratio "n = Es/Ec"	7.19	
Floor Finish	3.5	kN/m^2	**Provided Reinforcement**		
Wall load	0	kN/m^2	Top layer Nos.	6 Φ 10	mm
Total dead Load "U_{dl}"	7.1	kN/m^2	Bottom layer Nos.	0 Φ 12	mm
Live load	1	kN/m^2	Area of Top steel "$A_{s\ t}$"	468	mm^2
Total Ultimate Load "$Uu_{,tot}$"$_{(1.2D+1.6L)}$	10.12	kN/m^2	Area of Bottom steel "$A_{s\ b}$"	0	mm^2
Moment due to Overhang Point Load(0.125*0.15*24*0.6)	0.27	kNm	"d_b' = 10 mm		
Calculations					
Shear force Vu = U*le	**8.096**	**kN**			
Neg. Moment Mu = U*le^2/2	**3.51**	**kNm**			

Check for Section capacity								
Depth of comp. block "a" = $A_{s\ t}$*f_y/(0.85*f'_c*b)	6.61	mm						
Section capacity ΦMn = Φ*$A_{s\ t}$*f_y*(d-a/2)	**21.94**	**kNm**	**O.K**	>	**3.51**	**kNm**	(Mu)	
Shear requirements								
Shear capacity ΦVc = Φ*0.17*√f'c*b*d	**94.29**	**kN**	**O.K**	>	**8.096**	**kN**	(Vu)	
Serviceability requirements								
d' = c+d_b/2	26	mm						
B = b/(n*$A_{s\ t}$)	0.297							
r = (n-1)*$A_{s\ b}$/(n*$A_{s\ t}$)	0							
k_d = (√(2*d*B(1+r*d'/d)+(1+r)^2)-(1+r))/B	29.01	mm						
I_{cr} = bkd^3/3+n*$A_{s\ t}$*(d-k_d)^2+(n-1)*$A_{s\ b}$*(k_d-d'	3.915E+07	mm^4						
Calculate Gross moment of inertia								
Area of slab A = t*b	2E+05	mm^2						
Depth of neutral axis "y"	75	mm						
Distance of top fiber from N.A "y_t"	75	mm						
Gross moment of inertia "I_g"	2.813E+08	mm^4						
Allowable stress fr = 0.62√f'c	3.67	mm^2						
Cracking moment M_{cr} = fr*I_g/y_t	13.75	kNm						

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Service moments

Moment due to dead load "M_{dl}"	2.54	kNm	
Moment due to live load "Mll"	0.32	kNm	
Total Moment Ms = $M_{dl}+M_{ll}$	2.86	kNm	
Deflection due to dead + live loads			
R(dl+ll) = M_{cr}/Ms	1.000		
I_e(dl+ll) = $R^3*I_g+(1-R^3)*I_{cr}$	2.813E+08	mm^4	
δ = K*5/48*Ms*le^2/(Ec*I_e)	0.08	mm	**K = 3.2 (conservative)**
Deflection due to dead load			
R(dl) = M_{cr}/Md	5.411		
I_e(dl) = $R^3*I_g+(1-R^3)*I_{cr}$	3.839E+10	mm^4	
δ = K*5/48*Md*le^2/(Ec*I_e)	0.00	mm	

Deflection due to live load

Live load deflection = dl+ll deflection - dl deflection 0.08 mm

Allowable immediate deflection due to Live load = per ACI 318 Tab 9.5b I_e/360 **2.22 mm > 0.08 mm O.K**

Long term deflection

% of liveload sustained "SL"	50	%
M_{sus} = $M_{dl}+SL*M_{ll}$	2.70	kNm
R(sus) = M_{cr}/Msus	5.091	
I_e(sus) = $R^3*I_g+(1-R^3)*I_{cr}$	3.198E+10	
δ = K*5/48*Msus*le^2/(Ec*I_e)	0.00	mm

Magnification factor for creep and shrinkage

$\lambda_\Delta = \xi / (1+50*\rho')$ 2.000

Where:-

ξ = 2 for 5 years or more cl.9.5.2.5 ACI 318

$\rho' = A_{s\ b}/(b*d)$ 0

Total deflection = λ_Δ*sus deflection + ll deflection 0.079 mm

Allowable deflection = I_e/240 3.333 mm > 0.079 mm O.K

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

14. FOUNDATION DESIGN

Foundations have been designed based on the Soil Report Recommendations. The minimum Embedment Depth of Foundations has been considered as 1.2m & applicable SBC of 350 Kpa. (Refer Doc-MD-216-3000-EG-CV-RPT-1008 for Details)

TABLE 4: RECOMMENDED NET ALLOWABLE BEARING CAPACITIES

PRPJECT: MANSOURAH - MANSSARAH GOLD PROJECT
CLIENT: L & T CONSTRUCTION

Area	Borehole No.	Proposed Structures	Foundation Type	Depth of Foundation below FGL	Footing Size (B x L)	Net Allowable Bearing Capacity Based on		Recommended Net Allowable Bearing Capacity	Estimated Total Settlement	Vertical Modulus of Subgrade Reaction (Ks)	Horizontal Modulus of Subgrade Reaction (Kh)
						Settlement	Shear Failure				
				(m)	(m)	(kN/m^2)	(kN/m^2)	(kN/m^2)	(mm)	($kN/m^2/m$)	(kN/m^3)
Village Area	BH-21	Village	Isolated Square	1.20	1.0 x 1.0	1000	350	350	25	42,000	32,200
				1.50		1100	500	500	25	60,000	38,900
				1.20	2.0 x 2.0	700	580	580	25	69,600	37,500
				1.50		850	750	750	25	90,000	44,200
				1.20	3.0 x 3.0	600	750	600	25	72,000	42,800
				1.50		700	930	700	25	84,000	49,500
				1.20	4.0 x 4.0	550	910	550	25	66,000	48,100
				1.50		625	1090	625	25	75,000	54,800
				1.20	5.0 x 5.0	475	1060	475	25	57,000	53,400
				1.50		550	1240	550	25	66,000	60,200

Refer Attachment- II for Foundation Design calculations

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

REFERENCES

1.)	BUILDING PLANS	MD-216-1800-EG-AR-GAD-3152
2.)	OVERALL LAYOUT PLAN	MD-216-1800-EG-CV-GAD-1565
3.)	LAYOUT & DETAILS OF FOUNDATION, COLUMN & GRADE BEAM	MD-216-1800-EG-CV-GAD-1546
4.)	LAYOUT & DETAILS OF ROOF & GRADE SLAB	MD-216-1800-EG-CV-GAD-1569
5.)	GEOTECHNICAL REPORT	MD-216-3000-EG-CV-RPT-1008
6.)	STRUCTURAL DESIGN CRITERIA	MD-216-0000-EG-ST-CRT-0001

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

ATTACHMENT 1 STAAD INPUT DATA

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 18-Oct-19
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 0 0; 4 0 3.5 0; 23 0 -1.2 0; 24 5.25 0 0; 25 5.25 3.5 0; 26 5.25 -1.2 0;
27 10.5 0 0; 28 10.5 3.5 0; 29 10.5 -1.2 0; 30 15.75 0 0; 31 15.75 3.5 0;
32 15.75 -1.2 0; 33 0 0 4.25; 34 0 3.5 4.25; 35 0 -1.2 4.25; 37 5.25 3.5 4.25;
38 5.25 -1.2 4.25; 40 10.5 3.5 4.25; 41 10.5 -1.2 4.25; 42 15.75 0 4.25;
43 15.75 3.5 4.25; 44 15.75 -1.2 4.25; 45 0 0 14.35; 46 0 3.5 14.35;
47 0 -1.2 14.35; 54 15.75 0 14.35; 55 15.75 3.5 14.35; 56 15.75 -1.2 14.35;
57 5.25 0 14.35; 58 5.25 3.5 14.35; 59 5.25 -1.2 14.35; 60 10.25 0 14.35;
61 10.25 3.5 14.35; 62 10.25 -1.2 14.35; 63 0 0 9.35; 64 0 -1.2 9.35;
65 15.75 0 9.35; 66 15.75 -1.2 9.35; 71 0 3.5 9.35; 72 15.75 3.5 9.35;
MEMBER INCIDENCES
3 1 4; 31 1 23; 32 1 24; 33 4 25; 35 24 25; 36 24 26; 37 24 27; 38 25 28;
40 27 28; 41 27 29; 42 27 30; 43 28 31; 45 30 31; 46 30 32; 47 1 33; 48 4 34;
56 30 42; 57 31 43; 59 33 34; 60 33 35; 63 37 38; 67 40 41; 71 42 43; 72 42 44;
73 33 63; 74 34 71; 82 42 65; 83 43 72; 85 45 46; 86 45 47; 91 54 55; 92 54 56;
93 57 58; 94 57 59; 95 60 61; 96 60 62; 97 46 58; 98 58 61; 99 61 55;
100 45 57; 101 57 60; 102 60 54; 103 63 45; 104 63 64; 105 65 54; 106 65 66;
116 71 46; 117 63 71; 118 72 55; 119 65 72; 120 34 37; 121 37 40; 122 40 43;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17184e+007
POISSON 0.17
DENSITY 23.6158
ALPHA 5e-006
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27578.9
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
59 71 93 95 117 119 PRIS YD 0.5 ZD 0.25
MEMBER PROPERTY AMERICAN
33 38 43 PRIS YD 0.4 ZD 0.25
97 TO 99 120 TO 122 PRIS YD 0.5 ZD 0.25
32 42 47 56 73 82 100 102 103 105 PRIS YD 0.5 ZD 0.25
48 57 74 83 116 118 PRIS YD 0.4 ZD 0.25
MEMBER PROPERTY AMERICAN
63 67 PRIS YD 0.5 ZD 0.25
MEMBER PROPERTY AMERICAN
37 101 PRIS YD 0.5 ZD 0.25
MEMBER PROPERTY AMERICAN
85 91 PRIS YD 0.5 ZD 0.25
MEMBER PROPERTY AMERICAN
60 72 86 92 94 96 104 106 PRIS YD 0.5 ZD 0.25
MEMBER PROPERTY
3 31 35 36 40 41 45 46 PRIS YD 0.45 ZD 0.25
CONSTANTS
BETA 90 MEMB 59 60 71 72 104 106 117 119
MATERIAL CONCRETE ALL
SUPPORTS
23 26 29 32 35 38 41 44 47 56 59 62 64 66 FIXED
**************************************************************************
MEMBER CRACKED
3 31 35 36 40 41 45 46 59 60 63 67 71 72 85 86 91 TO 96 104 106
```

```
REDUCTION RIY 0.700000 RIZ 0.700000
32 33 37 38 42 43 47 48 56 57 73 74 82 83 97 TO 103 105 116 118 120 TO 121 -
122 REDUCTION RIY 0.350000 RIZ 0.350000
MEMBER OFFSET
33 38 43 97 TO 99 120 TO 122 START 0.25 0 0
33 38 43 97 TO 99 120 TO 122 END -0.25 0 0
48 74 START 0 0 0.125
48 116 END 0 0 -0.125
DEFINE IBC 2003
SDS 0.106 SD1 0.032 S1 0.02 IE 1 RX 2.5 RZ 2.5 SCLASS 4
SELFWEIGHT 1
MEMBER WEIGHT
97 TO 99 UNI 36.25
32 37 42 47 56 73 82 100 TO 103 105 UNI 12
33 38 43 UNI 12.36
DEFINE WIND LOAD
TYPE 1 WIND X WINDWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.000 M 20.000 M 15.000 M 2.000 0.010 0 -
0 0 0 0 1.030 1.000 1.000 0.850 0 -
1 0 0 0.850 0.800 -0.550
!> END GENERATED DATA BLOCK
INT 1.59699 1.59699 HEIG  0  4.572
TYPE 2 WIND X LEEWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.000 M 16.000 M 16.000 M 2.000 0.010 1 -
0 0 1 0 1.030 1.000 1.000 0.850 0 -
1 0 0 0.850 -0.500 0.550
!> END GENERATED DATA BLOCK
INT -1.26591 -1.26591 HEIG  0  4
TYPE 3 SIDEWARD X
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.000 M 20.000 M 15.000 M 2.000 0.010 2 -
0 0 1 0 1.030 1.000 1.000 0.850 0 -
1 0 0 0.850 -0.700 0.550
!> END GENERATED DATA BLOCK
INT -1.48663 -1.48663 HEIG  0  4
TYPE 4 WIND Z WINDWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.000 M 15.000 M 20.000 M 2.000 0.010 0 -
0 0 1 0 1.030 1.000 1.000 0.850 0 -
1 0 0 0.850 0.800 -0.550
!> END GENERATED DATA BLOCK
INT 1.59699 1.59699 HEIG  0  4.572
TYPE 5 WIND Z LEEWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.000 M 15.000 M 20.000 M 2.000 0.010 1 -
0 0 1 0 1.030 1.000 1.000 0.850 0 -
1 0 0 0.850 -0.500 0.550
!> END GENERATED DATA BLOCK
INT -1.26591 -1.26591 HEIG  0  4
TYPE 6 SIDEWARD Z
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
```

```
1 4.000 M 15.000 M 20.000 M 2.000 0.010 2 -
0 0 1 0 1.030 1.000 1.000 0.850 0 -
1 0 0 0.850 -0.700 0.550
!> END GENERATED DATA BLOCK
INT -1.48663 -1.48663 HEIG  0  4
LOAD 1 LOADTYPE Seismic  TITLE EQK LOAD Z
IBC LOAD Z 1
PERFORM ANALYSIS
CHANGE
LOAD 2 LOADTYPE Seismic  TITLE EQK LOAD X
IBC LOAD X 1
PERFORM ANALYSIS
CHANGE
LOAD 3 LOADTYPE Dead  TITLE DL
SELFWEIGHT Y -1
MEMBER LOAD
120 TO 122 UNI GY -48.62
97 TO 99 UNI GY -36.25
48 74 116 UMOM GZ 1
57 83 118 UMOM GZ -1
33 38 43 UMOM GX -1
97 TO 99 UMOM GX 1
32 37 42 47 56 73 82 100 TO 103 105 UNI GY -12
33 38 43 UNI GY -12.36
LOAD 4 LOADTYPE Live  TITLE LL
MEMBER LOAD
48 74 116 UMOM GZ 0.18
57 83 118 UMOM GZ -0.18
33 38 43 UMOM GX -0.18
97 TO 99 UMOM GX 0.18
33 38 43 UNI GY -1.06
120 TO 122 UNI GY -3.56
97 TO 99 UNI GY -2.5
LOAD 5 LOADTYPE Wind  TITLE WL+X
WIND LOAD X 1 TYPE 1 YR 0 4
WIND LOAD -X 1 TYPE 2 YR 0 4
WIND LOAD -Z 1 TYPE 3 YR 0 4
WIND LOAD -Z -1 TYPE 3 YR 0 4
LOAD 6 LOADTYPE Wind  TITLE WL+Z
WIND LOAD Z 1 TYPE 4 YR 0 4
WIND LOAD -Z 1 TYPE 5 YR 0 4
WIND LOAD -X 1 TYPE 6 YR 0 4
WIND LOAD -X -1 TYPE 6 YR 0 4
LOAD 7 LOADTYPE Wind  TITLE WL-X
WIND LOAD X -1 TYPE 1 YR 0 4
WIND LOAD -X -1 TYPE 2 YR 0 4
WIND LOAD -Z 1 TYPE 3 YR 0 4
WIND LOAD -Z -1 TYPE 3 YR 0 4
LOAD 8 LOADTYPE Wind  TITLE WL-Z
WIND LOAD Z -1 TYPE 4 YR 0 4
WIND LOAD -Z -1 TYPE 5 YR 0 4
WIND LOAD -X 1 TYPE 6 YR 0 4
WIND LOAD -X -1 TYPE 6 YR 0 4
LOAD 9 LOADTYPE Temperature  TITLE TEMP
TEMPERATURE LOAD
3 33 35 38 40 43 45 48 57 59 71 74 83 85 91 93 95 97 TO 99 116 TO 121 -
122 TEMP 25 25 25
LOAD COMB 10 SLS: (DL)
3 1.0
```

LOAD COMB 11 SLS: (DL+T)
3 1.0
LOAD COMB 12 SLS: (D+LR)
3 1.0 4 1.0
LOAD COMB 13 SLS: (D+0.75LR)
3 1.0 4 0.75
LOAD COMB 14 SLS: (D+0.7EZ)
3 1.0 1 0.7
LOAD COMB 15 SLS: (D+0.7EX)
3 1.0 2 0.7
LOAD COMB 16 SLS: (D+0.75WX1+0.75LR)
3 1.0 5 0.75 4 0.75
LOAD COMB 17 SLS: (D+0.75WZ1+0.75LR)
3 1.0 6 0.75 4 0.75
LOAD COMB 18 SLS: (0.6D+WX1)
3 0.6 5 1.0
LOAD COMB 19 SLS: (0.6D+WZ1)
3 0.6 6 1.0
LOAD COMB 20 SLS: (0.6D+0.7EX)
3 0.6 2 0.7
LOAD COMB 21 SLS: (0.6D+0.7EZ)
3 0.6 1 0.7
LOAD COMB 22 SLS: (D+WX1)
3 1.0 5 1.0
LOAD COMB 23 SLS: (D+WZ1)
3 1.0 6 1.0
LOAD COMB 24 ULS: (1.4 DL)
3 1.4
LOAD COMB 25 ULS: (1.4D+1.4T+0.5LR)
3 1.4 4 0.5
LOAD COMB 26 ULS: (1.4D+1.4T+0.5LR)
3 1.4 4 0.5
LOAD COMB 27 ULS: (1.2D+0.80WX1+1.6LR)
3 1.2 5 0.8 4 1.6
LOAD COMB 28 ULS: (1.2D+1.6WZ1+0.5LR)
3 1.2 6 1.6 4 0.5
LOAD COMB 29 ULS: (1.2D+1.6WX1+0.5LR)
3 1.2 4 0.5 5 1.6
LOAD COMB 30 ULS: (1.2D+1.6WZ1+0.5LR)
3 1.2 4 0.5 6 1.6
LOAD COMB 31 ULS: (1.2D+EX)
3 1.2 2 1.0
LOAD COMB 32 ULS: (1.2D+EZ)
3 1.2 1 1.0
LOAD COMB 33 ULS: (0.90D+1.6WX1)
3 0.9 5 1.6
LOAD COMB 34 ULS: (0.90D+1.6WZ1)
3 0.9 6 1.6
LOAD COMB 35 ULS: (0.90D+EZ)
3 0.9 1 1.0
LOAD COMB 36 ULS: (0.90D+EX)
3 0.9 2 1.0
LOAD COMB 39 SLS: (D+0.75WX2+0.75LR)
3 1.0 7 0.75 4 0.75
LOAD COMB 40 SLS: (D+0.75WZ2+0.75LR)
3 1.0 8 0.75 4 0.75
LOAD COMB 41 SLS: (0.6D+WX2)
3 0.6 7 1.0
LOAD COMB 42 SLS: (0.6D+WZ2)

```
3 0.6 8 1.0
LOAD COMB 45 SLS: (D+WX2)
3 1.0 7 1.0
LOAD COMB 46 SLS: (D+WZ2)
3 1.0 8 1.0
LOAD COMB 47 ULS: (1.2D+0.80WX2+1.6LR)
3 1.2 7 0.8 4 1.6
LOAD COMB 48 ULS: (1.2D+1.6WZ2+0.5LR)
3 1.2 8 1.6 4 0.5
LOAD COMB 49 ULS: (1.2D+1.6WX2+0.5LR)
3 1.2 4 0.5 7 1.6
LOAD COMB 50 ULS: (1.2D+1.6WZ2+0.5LR)
3 1.2 4 0.5 8 1.6
LOAD COMB 53 ULS: (0.90D+1.6WX2)
3 0.9 7 1.6
LOAD COMB 54 ULS: (0.90D+1.6WZ2)
3 0.9 8 1.6
PERFORM ANALYSIS
START CONCRETE DESIGN
CODE ACI
FC 35000 ALL
CLB 0.04 ALL
CLS 0.04 ALL
CLT 0.04 ALL
MINSEC 10 ALL
SFACE 0.5 MEMB 97 TO 99 120 TO 122
EFACE 0.5 MEMB 97 TO 99 120 TO 122
SFACE 0.4 MEMB 33 38 43
EFACE 0.4 MEMB 33 38 43
MAXMAIN 16 MEMB 32 37 42 47 56 73 82 100 TO 103 105
MINMAIN 16 MEMB 32 33 37 38 42 43 47 56 73 82 97 TO 103 105
MAXMAIN 20 MEMB 3 31 33 35 36 38 40 41 43 45 46 48 57 59 60 63 67 71 72 74 -
83 85 86 91 TO 99 104 106 116 118
MINMAIN 20 MEMB 3 31 35 36 40 41 45 46 48 57 59 60 63 67 71 72 74 83 85 86 -
91 TO 96 104 106 116 118
DESIGN BEAM 32 33 37 38 42 43 47 48 56 57 73 74 82 83 97 TO 103 105 116 118 -
120 TO 122
DESIGN COLUMN 3 31 35 36 40 41 45 46 59 60 63 67 71 72 85 86 91 TO 96 104 -
106 117 119
END CONCRETE DESIGN
FINISH
FINISH
```

ATTACHMENT 2 FOUNDATION DESIGN CALCULATIONS

PROPRIETARY & CONFIDENTIAL

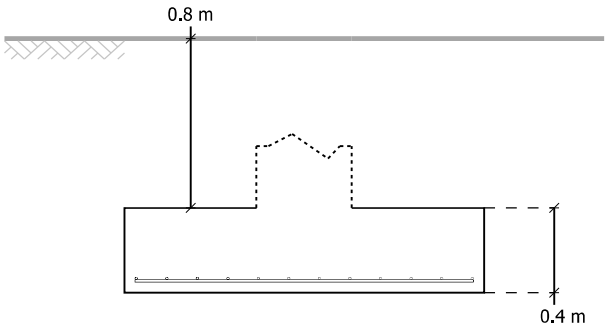
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

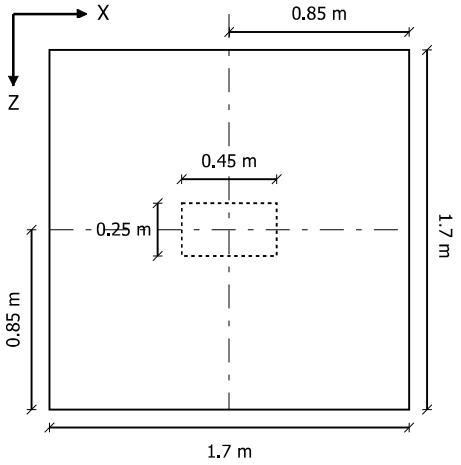
FOUNDATION-F4 ALONG GRID 1 & B

Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
29	1	1.700m	1.700m	0.400m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
29	12 - ϕ12	12 - ϕ12	N/A	N/A	N/A	N/A



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 400.000mm
Footing Length - X (Fl) : 1700.000mm
Footing Width - Z (Fw) : 1700.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.450m

Column Width - Z (B_{col}) : 0.250m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 125.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 800.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.530
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 1.700m

Initial Width (W_o) = 1.700m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
10	SLS: (DL)	1.00	1.00	1.00
11	SLS: (DL+T)	1.00	1.00	1.00
12	SLS: (D+LR)	1.00	1.00	1.00
13	SLS: (D+0.75LR)	1.00	1.00	1.00
14	SLS: (D+0.7EZ)	1.00	1.00	1.00
15	SLS: (D+0.7EX)	1.00	1.00	1.00
16	SLS: (D+0.75WX1+0.75LR)	1.00	1.00	1.00
17	SLS: (D+0.75WZ1+0.75LR)	1.00	1.00	1.00
18	SLS: (0.6D+WX1)	1.00	1.00	1.00
19	SLS: (0.6D+WZ1)	1.00	1.00	1.00
20	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
21	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
22	SLS: (D+WX1)	1.00	1.00	1.00
23	SLS: (D+WZ1)	1.00	1.00	1.00
39	SLS: (D+0.75WX2+0.75LR)	1.00	1.00	1.00
40	SLS: (D+0.75WZ2+0.75LR)	1.00	1.00	1.00
41	SLS: (0.6D+WX2)	1.00	1.00	1.00
42	SLS: (0.6D+WZ2)	1.00	1.00	1.00
45	SLS: (D+WX2)	1.00	1.00	1.00
46	SLS: (D+WZ2)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
24	ULS: (1.4 DL)	1.00	1.00	1.00
25	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
26	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+0.80WX1+1.6LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+1.6WX1+0.5LR)	1.00	1.00	1.00
30	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
31	ULS: (1.2D+EX)	1.00	1.00	1.00
32	ULS: (1.2D+EZ)	1.00	1.00	1.00
33	ULS: (0.90D+1.6WX1)	1.00	1.00	1.00
34	ULS: (0.90D+1.6WZ1)	1.00	1.00	1.00
35	ULS: (0.90D+EZ)	1.00	1.00	1.00
36	ULS: (0.90D+EX)	1.00	1.00	1.00
47	ULS: (1.2D+0.80WX2+1.6LR)	1.00	1.00	1.00
48	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
49	ULS: (1.2D+1.6WX2+0.5LR)	1.00	1.00	1.00
50	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
53	ULS: (0.90D+1.6WX2)	1.00	1.00	1.00
54	ULS: (0.90D+1.6WZ2)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
10	150.637	0.650	0.327	-0.186	-0.212
11	150.637	0.650	0.327	-0.186	-0.212
12	156.874	0.931	0.388	-0.342	-0.346
13	155.314	0.860	0.373	-0.303	-0.313
14	150.632	0.656	2.027	3.262	-0.218
15	150.443	4.010	0.371	-0.115	-3.779
16	155.025	6.099	-16.170	-30.293	-5.582
17	155.246	1.241	18.423	33.043	-0.596
18	89.996	7.375	-21.861	-40.099	-7.154
19	90.291	0.897	24.264	44.349	-0.505

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
20	90.189	3.750	0.240	-0.040	-3.694
21	90.377	0.396	1.896	3.336	-0.133
22	150.251	7.635	-21.730	-40.173	-7.239
23	150.546	1.157	24.395	44.275	-0.590
39	155.621	-4.507	-16.368	-30.610	5.082
40	155.251	1.204	-14.059	-27.174	-0.543
41	90.791	-6.767	-22.125	-40.521	7.066
42	90.297	0.848	-19.046	-35.939	-0.435
45	151.045	-6.507	-21.994	-40.595	6.981
46	150.552	1.108	-18.915	-36.014	-0.520

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
24	210.891	0.910	0.458	-0.260	-0.297
25	214.010	1.050	0.489	-0.338	-0.364
26	214.010	1.050	0.489	-0.338	-0.364
27	190.434	6.817	-17.156	-32.462	-6.090
28	183.737	1.732	38.931	70.836	-0.926
29	183.265	12.096	-34.869	-64.280	-11.564
30	183.737	1.732	38.931	70.836	-0.926
31	180.488	5.579	0.454	-0.122	-5.350
32	180.757	0.789	2.821	4.702	-0.263
33	134.956	11.761	-34.997	-64.147	-11.433
34	135.428	1.397	38.802	70.970	-0.795
35	135.566	0.594	2.723	4.758	-0.200
36	135.297	5.384	0.356	-0.066	-5.287
47	191.070	-4.496	-17.367	-32.800	5.285
48	183.746	1.654	-30.364	-57.625	-0.814
49	184.536	-10.530	-35.291	-64.956	11.187
50	183.746	1.654	-30.364	-57.625	-0.814
53	136.227	-10.865	-35.419	-64.822	11.318
54	135.437	1.318	-30.493	-57.491	-0.683

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 2.890 m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 1.149 m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 1.700 m Governing Load Case : # 10

Width (W_2) = 1.700 m Governing Load Case : # 10

Depth (D_2) = 0.400 m Governing Load Case : # 49

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 2.890 m^2

Final Soil Height = 0.800 m

Footing Self Weight = 27.743 kN

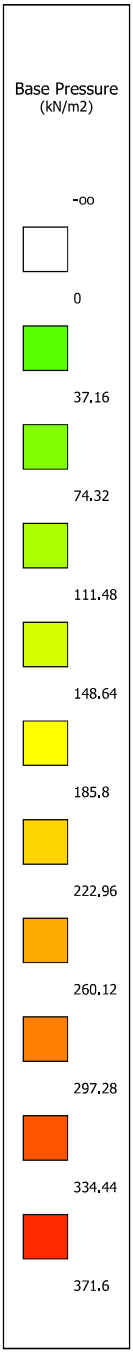
Gross Soil Bearing Capacity = 371.60 kN/m2

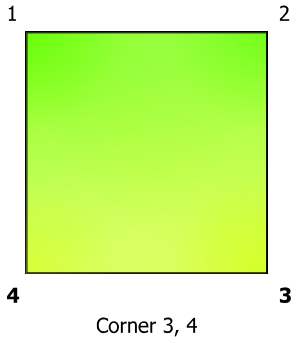
Soil Weight On Top Of Footing = 39.995 kN

Pressures at 4 Corners

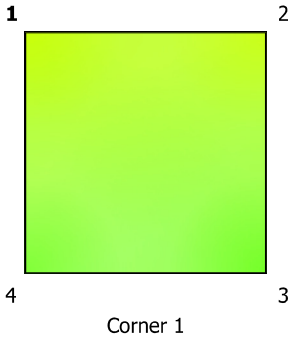
Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner

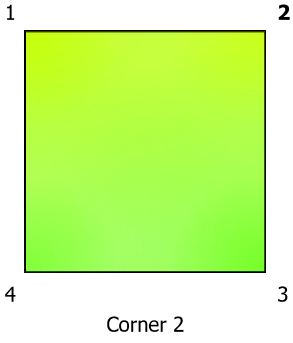




Corner 3, 4

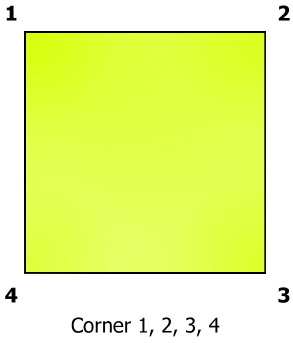


Corner 1



Corner 2

Governing Base Pressure Contour



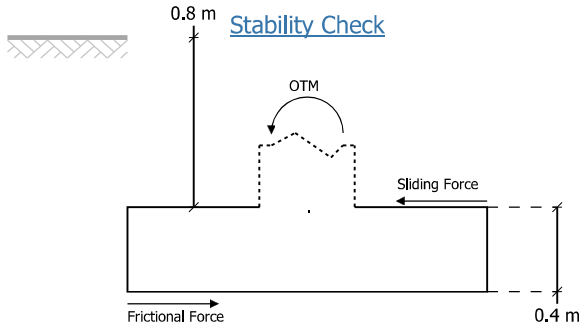
Corner 1, 2, 3, 4

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
45	**147.7286**	124.3211	3.6782	27.0857	0.000
22	122.5361	**147.6755**	28.3209	3.1815	0.000
23	8.2570	10.8279	**142.8044**	140.2334	0.000
23	8.2570	10.8279	142.8044	**140.2334**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
45	**147.7286**	124.3211	3.6782	27.0857
22	122.5361	**147.6755**	28.3209	3.1815
23	8.2570	10.8279	**142.8044**	140.2334
23	8.2570	10.8279	142.8044	**140.2334**



Stability Check

-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
10	178.100	353.625	224.952	1.500	3386.552	393.526	1.500
11	178.100	353.625	224.952	1.500	3386.552	393.526	1.500
12	127.926	306.827	166.982	1.500	1027.609	265.955	1.500
13	137.400	317.099	178.295	1.500	1238.839	288.876	1.500
14	176.425	57.101	76.829	1.500	45.824	386.553	1.500
15	28.840	312.100	40.613	1.500	5591.241	34.476	1.500
16	19.358	7.301	9.661	1.500	5.178	23.620	1.500
17	95.237	6.415	9.051	1.500	4.715	173.659	1.500
18	11.336	3.824	5.124	1.500	2.765	13.283	1.500
19	93.343	3.452	4.878	1.500	2.503	155.665	1.500
20	22.322	349.348	31.504	1.500	2441.535	25.870	1.500
21	211.584	44.200	61.187	1.500	33.068	461.177	1.500
22	15.133	5.317	7.094	1.500	3.812	18.016	1.500
23	99.972	4.742	6.699	1.500	3.452	176.400	1.500
39	26.266	7.232	9.861	1.500	5.136	27.596	1.500
40	98.138	8.406	11.845	1.500	5.809	185.036	1.500
41	12.417	3.798	5.136	1.500	2.749	13.803	1.500
42	98.725	4.398	6.213	1.500	3.107	173.669	1.500
45	17.821	5.272	7.150	1.500	3.785	19.419	1.500
46	104.384	6.116	8.635	1.500	4.280	192.801	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 18

Governing Disturbing Force : 7.375kN

Governing Restoring Force : 83.599kN

Minimum Sliding Ratio for the Critical Load Case : 11.336

Critical Load Case for Overturning about X-Direction : 19

Governing Overturning Moment : 54.054kNm

Governing Resisting Moment : 135.317kNm

Minimum Overturning Ratio for the Critical Load Case : 2.503

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 19

Governing Disturbing Force : 24.264kN

Governing Restoring Force : 83.755kN

Minimum Sliding Ratio for the Critical Load Case : 3.452

Critical Load Case for Overturning about Z-Direction : 18

Governing Overturning Moment : -10.104kNm

Governing Resisting Moment : 134.209kNm

Minimum Overturning Ratio for the Critical Load Case : 13.283

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

Critical Load Case for Sliding along Resultant Direction : 19

Governing Disturbing Force : 24.280kN

Governing Restoring Force : 118.448kN

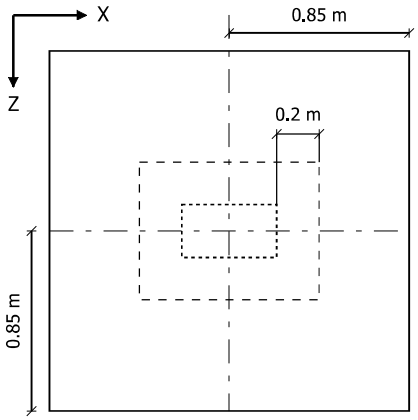
Minimum Sliding Ratio for the Critical Load Case : 4.878

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.400m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.344m

For rectangular column, β_c = B_{col} / D_{col} = 1.800

Effective depth, d_{eff}, increased until 0.75XV_c $\geq$ Punching Shear Force

Punching Shear Force, Vu = 235.767kN, Load Case # 25

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 2.776m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 1980.672kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 3263.468kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 1876.426kN

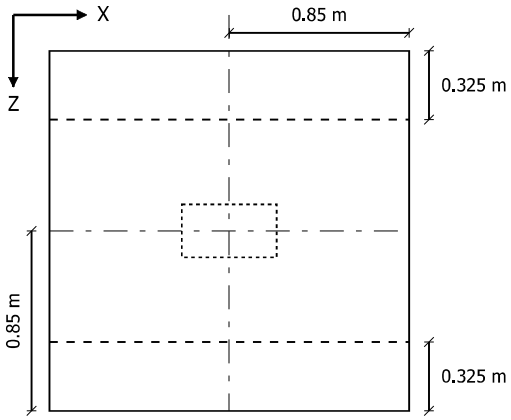
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1407.320kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 574.554kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{zd}$ = 0.381m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

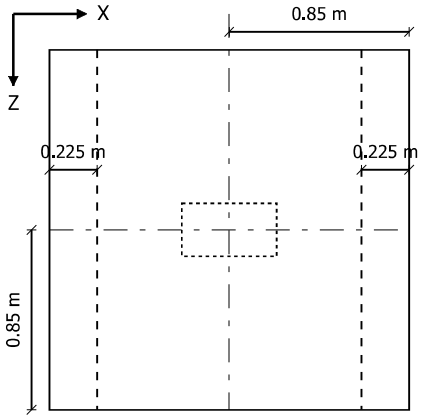
From above calculations, 0.75 X V_c = 430.916 kN

Critical load case for V_{ux} is # 28 $V_{ux} = V_{ux}|_{x=D_x}$ = 107.054 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 574.554 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.281 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

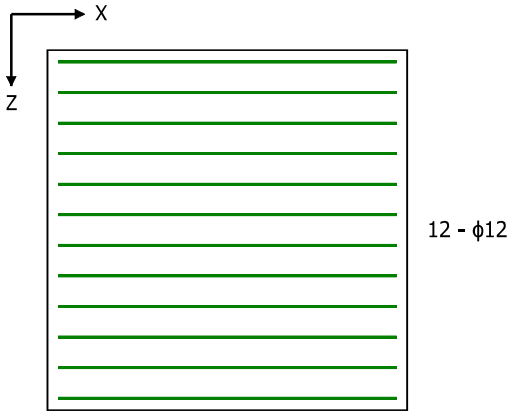
From above calculations, 0.75 X V_c = 430.916 kN

Critical load case for V_{uz} is # 49 $V_{uz} = V_{uz}|_{z=D_z} =$ 49.198 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 29

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.344 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488
From ACI Cl. 7.12.2, ρ_{min} =		0.00165
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	1.075 m
Ultimate moment,	$M_u\|_{z=D_z}$ =	34.998 kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	38.886 kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00046
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00040
Since	ρ= ρ_{min}	ρ_{min} Governs
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	1124.583 mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 125.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f'_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times (L - D_{col}) - C_{cover}$ = 0.575 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 12

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1357.159 mm2

d_{eff} = D - C_{cover} - 0.5 X (dia. of one bar) = 0.344 m

Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00232

From ACI Cl.7.6.1, minimum req'd clear distance between bars

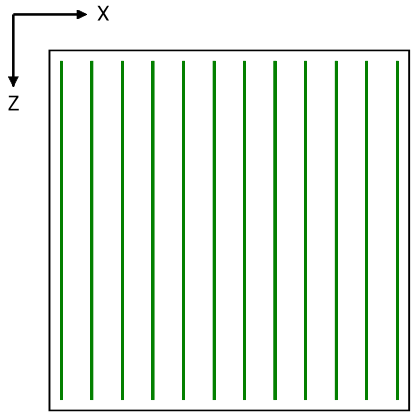
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.207

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



12 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 28

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} = 0.332 m

Factor β_1 from ACI Cl.10.2.7.3 = 0.796

From ACI Cl. 10.3.2, ρ_{bal} = $0.85 \times \beta_1 \times f'_c \times \left(\frac{600}{600 + f_y}\right)$ 0.03317

From ACI Cl. 10.3.3, ρ_{max} = $0.75 \times \rho_{bal}$ = 0.02488

From ACI Cl.7.12.2, ρ_{min} = 0.00170

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{\left(0.85 \times F_c'\right)}$ = 14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 0.975 m

Ultimate moment, $M_u\big|_{x=D_x}$ = 74.809 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 83.121 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00106

(Based on gross depth) ρ x d_{eff} / Depth = 0.00088

Since ρ= ρ_{min} ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 1155.119 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 125.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times \left(L - D_{col}\right) - C_{cover}$ = 0.675 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 12

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1357.159 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.332 m

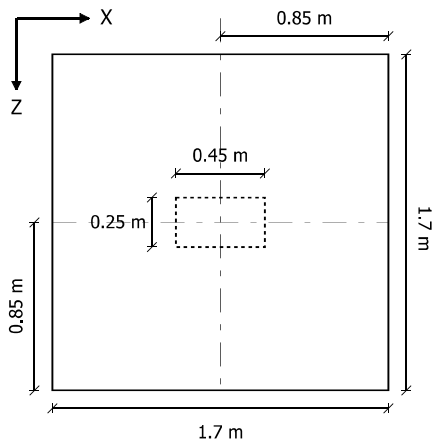
Reinforcement ratio, ρ = $\frac{A_{s_total}}{\left(d_{eff} \times W\right)}$ = 0.00240

From ACI Cl.7.6.1, minimum req'd clear distance between bars

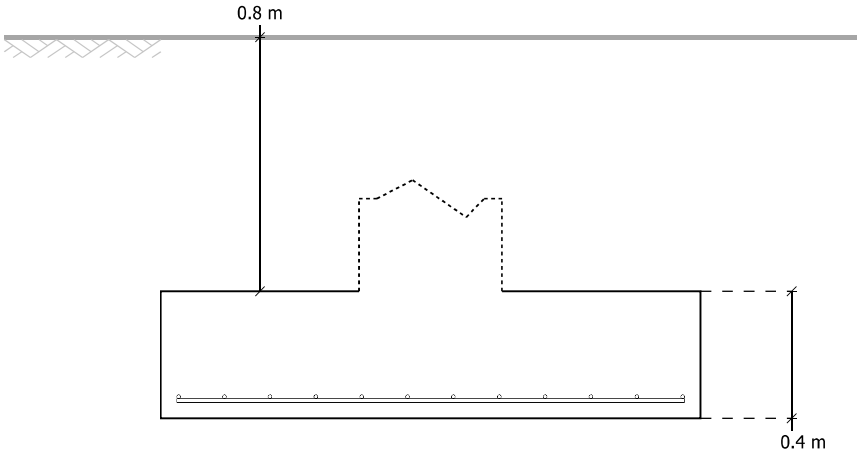
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.175

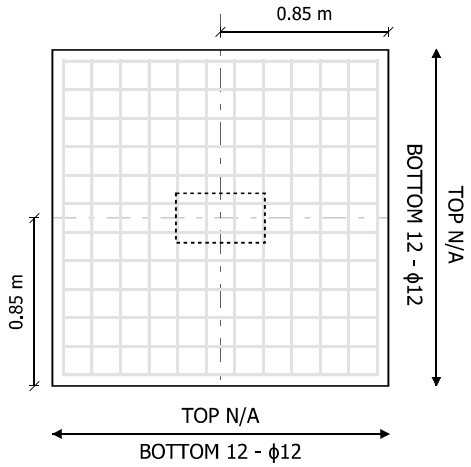
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

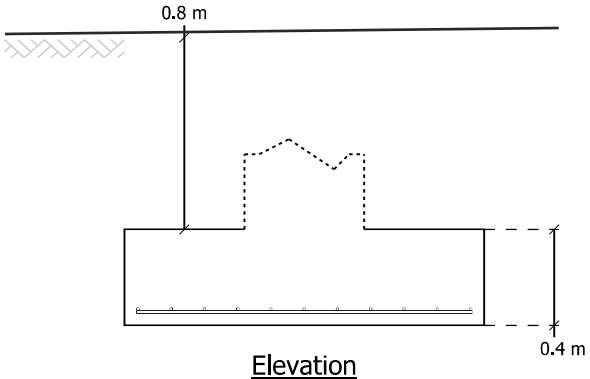
	Dimensions (m)			Reinforcement			
	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
Foundation	1.7 m	1.7 m	0.4 m	N/A	N/A	12 - ϕ12	12 - ϕ12
	l	b	h	Main Reinforcement		Trans Reinforcement	
Pedestal	N/A	N/A	N/A	N/A		N/A	

Print Calculation Sheet

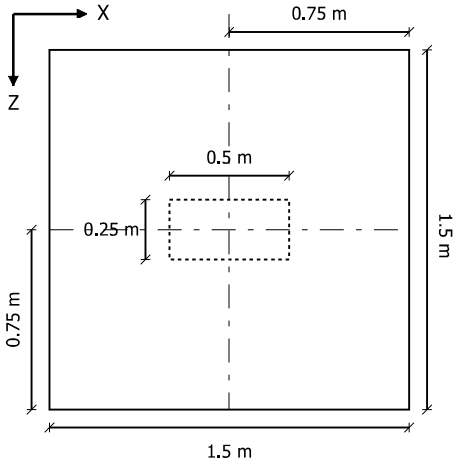
FOUNDATION-F3 ALONG GRID 2 & B-12

Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
38	1	1.500m	1.500m	0.400m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
38	11 - ϕ12	11 - ϕ12	N/A	N/A	N/A	N/A



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 400.000mm
Footing Length - X (Fl) : 1500.000mm
Footing Width - Z (Fw) : 1500.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.500m

Column Width - Z (B_{col}) : 0.250m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 125.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 800.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.530
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 1.500m

Initial Width (W_o) = 1.500m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
10	SLS: (DL)	1.00	1.00	1.00
11	SLS: (DL+T)	1.00	1.00	1.00
12	SLS: (D+LR)	1.00	1.00	1.00
13	SLS: (D+0.75LR)	1.00	1.00	1.00
14	SLS: (D+0.7EZ)	1.00	1.00	1.00
15	SLS: (D+0.7EX)	1.00	1.00	1.00
16	SLS: (D+0.75WX1+0.75LR)	1.00	1.00	1.00
17	SLS: (D+0.75WZ1+0.75LR)	1.00	1.00	1.00
18	SLS: (0.6D+WX1)	1.00	1.00	1.00
19	SLS: (0.6D+WZ1)	1.00	1.00	1.00
20	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
21	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
22	SLS: (D+WX1)	1.00	1.00	1.00
23	SLS: (D+WZ1)	1.00	1.00	1.00
39	SLS: (D+0.75WX2+0.75LR)	1.00	1.00	1.00
40	SLS: (D+0.75WZ2+0.75LR)	1.00	1.00	1.00
41	SLS: (0.6D+WX2)	1.00	1.00	1.00
42	SLS: (0.6D+WZ2)	1.00	1.00	1.00
45	SLS: (D+WX2)	1.00	1.00	1.00
46	SLS: (D+WZ2)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
24	ULS: (1.4 DL)	1.00	1.00	1.00
25	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
26	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+0.80WX1+1.6LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+1.6WX1+0.5LR)	1.00	1.00	1.00
30	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
31	ULS: (1.2D+EX)	1.00	1.00	1.00
32	ULS: (1.2D+EZ)	1.00	1.00	1.00
33	ULS: (0.90D+1.6WX1)	1.00	1.00	1.00
34	ULS: (0.90D+1.6WZ1)	1.00	1.00	1.00
35	ULS: (0.90D+EZ)	1.00	1.00	1.00
36	ULS: (0.90D+EX)	1.00	1.00	1.00
47	ULS: (1.2D+0.80WX2+1.6LR)	1.00	1.00	1.00
48	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
49	ULS: (1.2D+1.6WX2+0.5LR)	1.00	1.00	1.00
50	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
53	ULS: (0.90D+1.6WX2)	1.00	1.00	1.00
54	ULS: (0.90D+1.6WZ2)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
10	295.772	5.030	-0.010	-0.007	-7.469
11	295.772	5.030	-0.010	-0.007	-7.469
12	315.010	5.347	-0.003	0.018	-7.941
13	310.200	5.268	-0.005	0.012	-7.823
14	295.774	5.027	0.833	3.013	-7.459
15	295.473	10.193	-0.141	-0.359	-19.877
16	309.414	14.250	-0.252	-0.580	-29.436
17	310.194	5.145	0.482	1.902	-7.518
18	176.416	14.994	-0.335	-0.793	-33.299
19	177.456	2.854	0.644	2.517	-4.075

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
20	177.164	8.181	-0.137	-0.356	-16.890
21	177.465	3.015	0.837	3.016	-4.472
22	294.725	17.006	-0.339	-0.795	-36.286
23	295.765	4.866	0.640	2.514	-7.062
39	311.124	-3.535	0.345	1.030	13.387
40	309.998	5.160	0.320	0.720	-7.602
41	178.695	-8.719	0.461	1.354	23.798
42	177.193	2.874	0.428	0.941	-4.186
45	297.004	-6.707	0.457	1.351	20.811
46	295.502	4.886	0.424	0.938	-7.174

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
24	414.081	7.043	-0.014	-0.010	-10.456
25	423.700	7.201	-0.011	0.003	-10.692
26	423.700	7.201	-0.011	0.003	-10.692
27	384.868	16.124	-0.264	-0.599	-32.772
28	364.533	5.932	1.031	4.037	-8.548
29	362.869	25.356	-0.535	-1.257	-55.306
30	364.533	5.932	1.031	4.037	-8.548
31	354.499	13.412	-0.199	-0.511	-26.689
32	354.930	6.031	1.192	4.306	-8.949
33	264.518	23.688	-0.535	-1.268	-52.830
34	266.183	4.265	1.030	4.027	-6.072
35	266.198	4.522	1.195	4.308	-6.709
36	265.767	11.903	-0.196	-0.508	-24.449
47	386.692	-2.846	0.372	1.118	12.906
48	364.113	5.964	0.685	1.516	-8.727
49	366.516	-12.584	0.739	2.177	36.049
50	364.113	5.964	0.685	1.516	-8.727
53	268.166	-14.252	0.739	2.167	38.526
54	265.763	4.296	0.685	1.505	-6.250

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 2.250m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 1.407m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 1.500 m Governing Load Case : # 10

Width (W_2) = 1.500 m Governing Load Case : # 10

Depth (D_2) = 0.400 m Governing Load Case : # 29

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 2.250 m^2

Final Soil Height = 0.800 m

Footing Self Weight = 21.599 kN

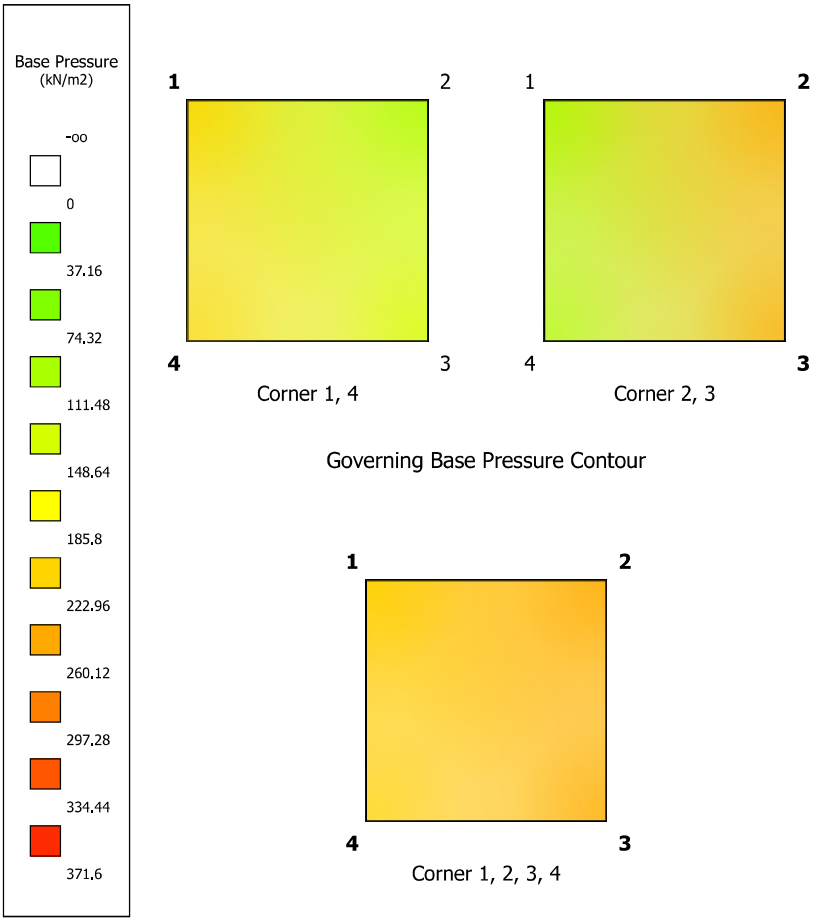
Gross Soil Bearing Capacity = 371.60 kN/m2

Soil Weight On Top Of Footing = 30.599 kN

Pressures at 4 Corners

Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner



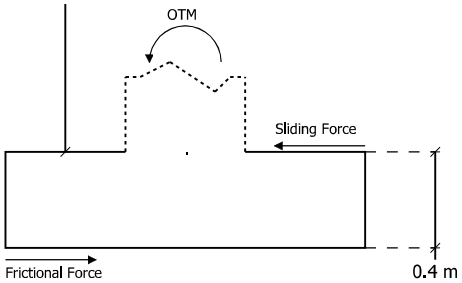
Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
45	**194.2406**	110.7065	116.1614	199.6955	0.000
22	79.2397	**232.4455**	229.1360	75.9302	0.000
22	79.2397	232.4455	**229.1360**	75.9302	0.000
45	194.2406	110.7065	116.1614	**199.6955**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
45	**194.2406**	110.7065	116.1614	199.6955
22	79.2397	**232.4455**	229.1360	75.9302
22	79.2397	232.4455	**229.1360**	75.9302
45	194.2406	110.7065	116.1614	**199.6955**

Stability Check



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
10	36.662	18296.076	51.848	1.500	23934.536	27.526	1.500
11	36.662	18296.076	51.848	1.500	23934.536	27.526	1.500
12	36.395	55845.090	51.470	1.500	16840.065	27.322	1.500
13	36.459	37418.898	51.560	1.500	28492.008	27.371	1.500
14	36.688	221.415	51.187	1.500	78.296	27.558	1.500
15	18.077	1308.259	25.563	1.500	630.989	10.885	1.500
16	13.449	761.828	19.017	1.500	400.142	7.719	1.500
17	37.331	398.427	52.563	1.500	130.216	28.383	1.500
18	8.081	362.084	11.425	1.500	186.172	4.363	1.500
19	42.647	189.133	58.835	1.500	62.455	33.018	1.500
20	14.859	888.500	21.011	1.500	421.549	8.532	1.500
21	40.376	145.431	55.019	1.500	51.711	30.336	1.500
22	10.812	542.925	15.287	1.500	280.625	6.038	1.500
23	37.898	288.373	53.139	1.500	94.591	28.968	1.500
39	54.478	557.462	76.679	1.500	234.077	18.410	1.500
40	37.204	599.452	52.513	1.500	321.415	28.105	1.500
41	14.036	265.244	19.821	1.500	113.210	6.346	1.500
42	42.307	284.210	59.179	1.500	155.662	32.243	1.500
45	27.596	404.691	38.936	1.500	171.369	11.148	1.500
46	37.717	434.891	53.141	1.500	236.406	28.568	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 18
Governing Disturbing Force : 14.994kN
Governing Restoring Force : 121.165kN
Minimum Sliding Ratio for the Critical Load Case : 8.081
Critical Load Case for Overturning about X-Direction : 21
Governing Overturning Moment : 3.351kNm
Governing Resisting Moment : 173.257kNm
Minimum Overturning Ratio for the Critical Load Case : 51.711

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 21
Governing Disturbing Force : 0.837kN
Governing Restoring Force : 121.722kN

Minimum Sliding Ratio for the Critical Load Case : 145.431

Critical Load Case for Overturning about Z-Direction : 18

Governing Overturning Moment : -39.296kNm

Governing Resisting Moment : 171.457kNm

Minimum Overturning Ratio for the Critical Load Case : 4.363

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

Critical Load Case for Sliding along Resultant Direction : 18

Governing Disturbing Force : 14.998kN

Governing Restoring Force : 171.354kN

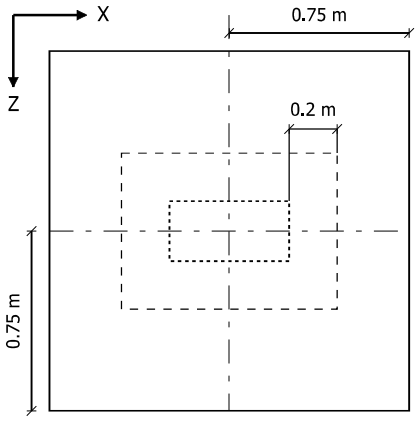
Minimum Sliding Ratio for the Critical Load Case : 11.425

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.400m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.344m

For rectangular column, β_c = B_{col} / D_{col} = 2.000

Effective depth, d_{eff}, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 369.860kN, Load Case # 25

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 2.876m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 1944.021kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 3297.265kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 1944.021kN

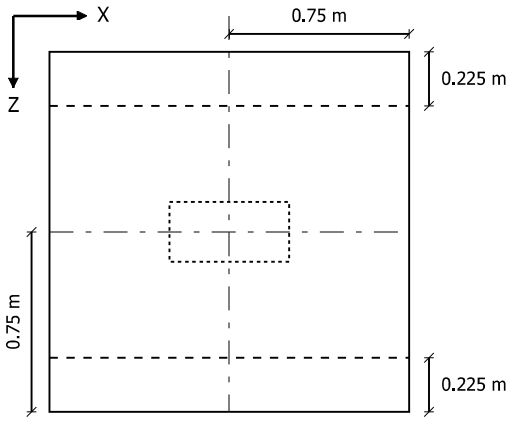
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1458.016kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 506.960kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{xd}$ = 0.281m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

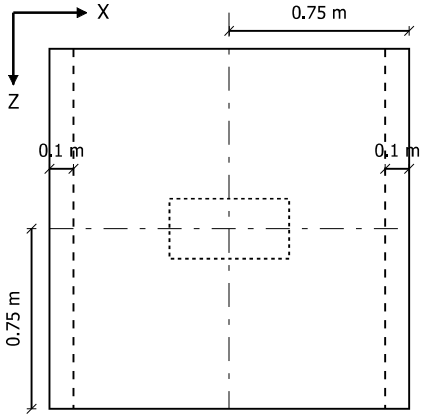
From above calculations, 0.75 X V_c = 380.220 kN

Critical load case for V_{ux} is # 25 $V_{ux} = V_{ux}|_{x=D_x}$ = 89.153 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 506.960 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.156 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

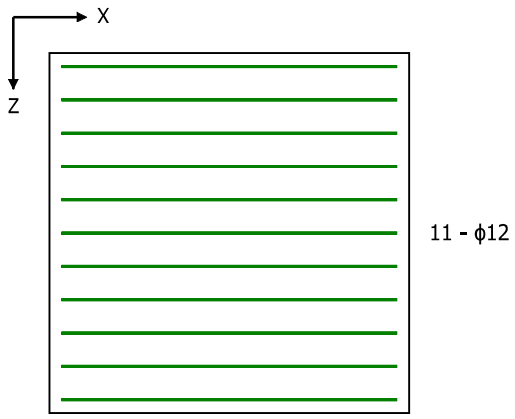
From above calculations, 0.75 X V_c = 380.220 kN

Critical load case for V_{uz} is # 29 $V_{uz} = V_{uz}|_{z=D_z}$ = 67.562 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 29

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.344 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488
From ACI Cl. 7.12.2, ρ_{min} =		0.00169
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	1.000 m
Ultimate moment,	$M_u\big\vert_{z=D_z}$ =	51.587 kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	57.319 kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00077
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00066
Since	ρ= ρ_{min}	ρ_{min} Governs
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	1014.827 mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 125.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = =0.410 m

$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}} =$

Available development length for bars, D_L = $0.5 \times (L - D_{col}) \cdot - C_{cover} =$ 0.450 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, $N_{bar} = \frac{A_s}{A_{bar}} =$ 11

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1244.062 mm2

d_{eff} = D - C_{cover} - 0.5 X (dia. of one bar) = 0.344 m

Reinforcement ratio, $\rho = \frac{A_{s_total}}{(d_{eff} \times W)} =$ 0.00241

From ACI Cl.7.6.1, minimum req'd clear distance between bars

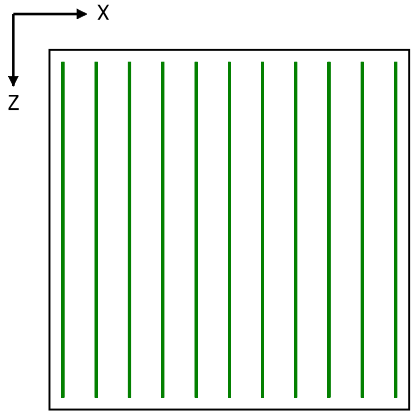
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.226

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



11 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 25

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} = 0.332 m

Factor β_1 from ACI Cl.10.2.7.3 = 0.796

From ACI Cl. 10.3.2, $\rho_{bal} = 0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$ 0.03317

From ACI Cl. 10.3.3, $\rho_{max} = 0.75 \times \rho_{bal} =$ 0.02488

From ACI Cl.7.12.2, ρ_{min} = 0.00174

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{(0.85 \times F_c')} =$ 14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 0.625 m

Ultimate moment, $M_u|_{x=D_x}$ = 61.965 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 68.850 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00100

(Based on gross depth) ρ x d_{eff} / Depth = 0.00083

Since $\rho = \rho_{min}$ ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 1041.114 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 125.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times (L - D_{col}) - C_{cover}$ = 0.575 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 11

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1244.062 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.332 m

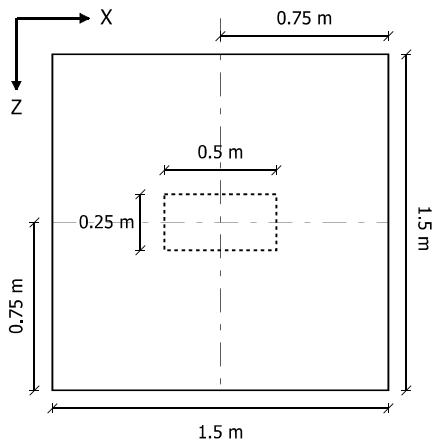
Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00250

From ACI Cl.7.6.1, minimum req'd clear distance between bars

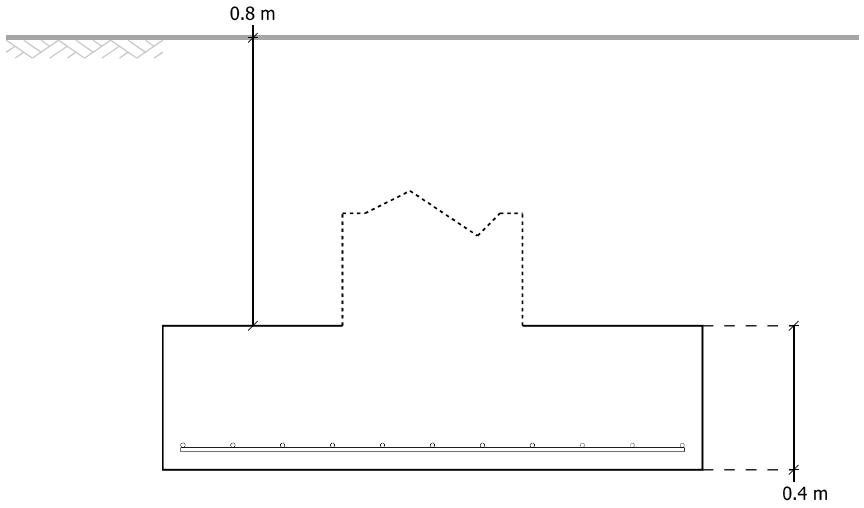
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.195

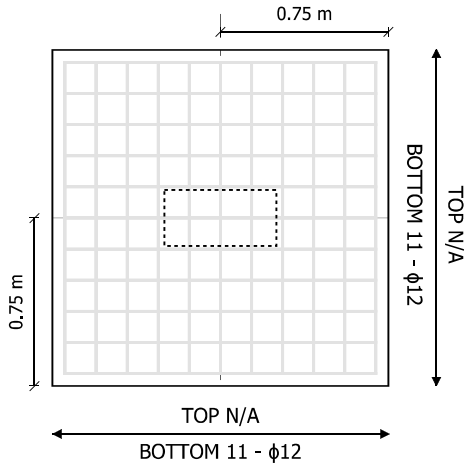
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

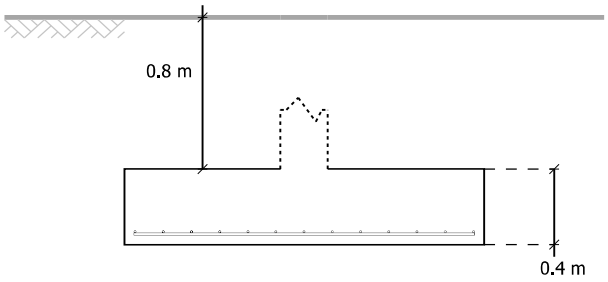
	Dimensions (m)			Reinforcement			
Foundation	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
	1.5 m	1.5 m	0.4 m	N/A	N/A	11 - ϕ12	11 - ϕ12
Pedestal	l	b	h	Main Reinforcement		Trans Reinforcement	
	N/A	N/A	N/A	N/A		N/A	

Print Calculation Sheet

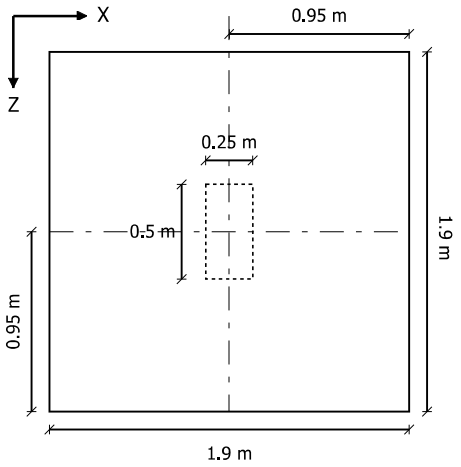
FOUNDATION-F2 ALONG GRID 3 & A

Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
64	1	1.900m	1.900m	0.400m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
64	13 - ϕ12	13 - ϕ12	N/A	N/A	N/A	N/A



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 400.000mm
Footing Length - X (Fl) : 1900.000mm
Footing Width - Z (Fw) : 1900.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.250m

Column Width - Z (B_{col}) : 0.500m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 125.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 800.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.530
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 1.900m

Initial Width (W_o) = 1.900m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
10	SLS: (DL)	1.00	1.00	1.00
11	SLS: (DL+T)	1.00	1.00	1.00
12	SLS: (D+LR)	1.00	1.00	1.00
13	SLS: (D+0.75LR)	1.00	1.00	1.00
14	SLS: (D+0.7EZ)	1.00	1.00	1.00
15	SLS: (D+0.7EX)	1.00	1.00	1.00
16	SLS: (D+0.75WX1+0.75LR)	1.00	1.00	1.00
17	SLS: (D+0.75WZ1+0.75LR)	1.00	1.00	1.00
18	SLS: (0.6D+WX1)	1.00	1.00	1.00
19	SLS: (0.6D+WZ1)	1.00	1.00	1.00
20	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
21	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
22	SLS: (D+WX1)	1.00	1.00	1.00
23	SLS: (D+WZ1)	1.00	1.00	1.00
39	SLS: (D+0.75WX2+0.75LR)	1.00	1.00	1.00
40	SLS: (D+0.75WZ2+0.75LR)	1.00	1.00	1.00
41	SLS: (0.6D+WX2)	1.00	1.00	1.00
42	SLS: (0.6D+WZ2)	1.00	1.00	1.00
45	SLS: (D+WX2)	1.00	1.00	1.00
46	SLS: (D+WZ2)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
24	ULS: (1.4 DL)	1.00	1.00	1.00
25	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
26	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+0.80WX1+1.6LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+1.6WX1+0.5LR)	1.00	1.00	1.00
30	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
31	ULS: (1.2D+EX)	1.00	1.00	1.00
32	ULS: (1.2D+EZ)	1.00	1.00	1.00
33	ULS: (0.90D+1.6WX1)	1.00	1.00	1.00
34	ULS: (0.90D+1.6WZ1)	1.00	1.00	1.00
35	ULS: (0.90D+EZ)	1.00	1.00	1.00
36	ULS: (0.90D+EX)	1.00	1.00	1.00
47	ULS: (1.2D+0.80WX2+1.6LR)	1.00	1.00	1.00
48	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
49	ULS: (1.2D+1.6WX2+0.5LR)	1.00	1.00	1.00
50	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
53	ULS: (0.90D+1.6WX2)	1.00	1.00	1.00
54	ULS: (0.90D+1.6WZ2)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
10	92.816	-0.391	-2.234	-0.996	0.659
11	92.816	-0.391	-2.234	-0.996	0.659
12	92.704	-0.320	-2.289	-1.020	0.670
13	92.732	-0.338	-2.275	-1.014	0.667
14	91.757	-0.400	2.540	7.802	0.673
15	92.778	3.006	-2.234	-1.013	-6.055
16	90.725	15.315	-1.554	-0.299	-21.671
17	90.159	-12.567	5.984	11.332	15.987
18	53.014	20.635	-0.380	0.356	-29.389
19	52.259	-16.540	9.672	15.864	20.821

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
20	55.652	3.162	-1.340	-0.615	-6.319
21	54.630	-0.243	3.433	8.200	0.409
22	90.140	20.479	-1.273	-0.043	-29.126
23	89.385	-16.696	8.779	15.465	21.085
39	90.887	-11.157	-1.556	-0.224	15.497
40	96.073	-14.773	-10.703	-13.323	20.135
41	53.230	-14.660	-0.382	0.456	20.168
42	60.144	-19.482	-12.578	-17.009	26.353
45	90.356	-14.816	-1.276	0.057	20.432
46	97.271	-19.638	-13.472	-17.408	26.616

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
24	129.943	-0.547	-3.127	-1.394	0.923
25	129.887	-0.512	-3.155	-1.406	0.928
26	129.887	-0.512	-3.155	-1.406	0.928
27	109.059	16.340	-2.000	-0.471	-23.020
28	105.834	-26.522	14.912	25.131	33.478
29	107.042	32.958	-1.171	0.318	-46.859
30	105.834	-26.522	14.912	25.131	33.478
31	111.325	4.383	-2.680	-1.220	-8.801
32	109.866	-0.482	4.139	11.373	0.810
33	79.253	33.040	-0.473	0.629	-47.062
34	78.045	-26.441	15.609	25.441	33.275
35	82.021	-0.364	4.809	11.672	0.612
36	83.480	4.500	-2.010	-0.921	-8.998
47	109.232	-11.896	-2.002	-0.391	16.626
48	118.451	-31.229	-20.689	-27.466	42.328
49	107.387	-23.514	-1.175	0.478	32.433
50	118.451	-31.229	-20.689	-27.466	42.328
53	79.599	-23.433	-0.477	0.789	32.229
54	90.662	-31.147	-19.991	-27.155	42.125

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 3.610m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 0.977m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 1.900 m Governing Load Case : # 10

Width (W_2) = 1.900 m Governing Load Case : # 10

Depth (D_2) = 0.400 m Governing Load Case : # 50

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 3.610 m^2

Final Soil Height = 0.800 m

Footing Self Weight = 34.655 kN

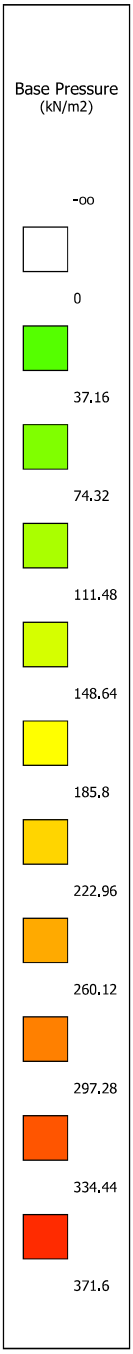
Gross Soil Bearing Capacity = 371.60 kN/m2

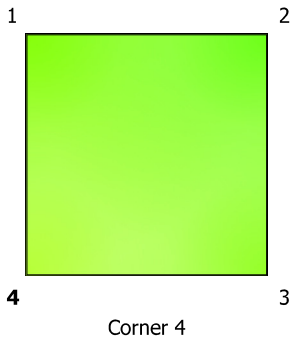
Soil Weight On Top Of Footing = 50.182 kN

Pressures at 4 Corners

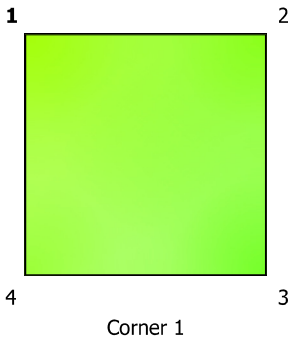
Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner

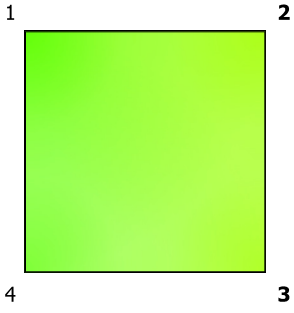




Corner 4

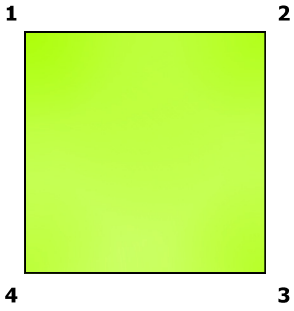


Corner 1



Corner 2, 3

Governing Base Pressure Contour



Corner 1, 2, 3, 4

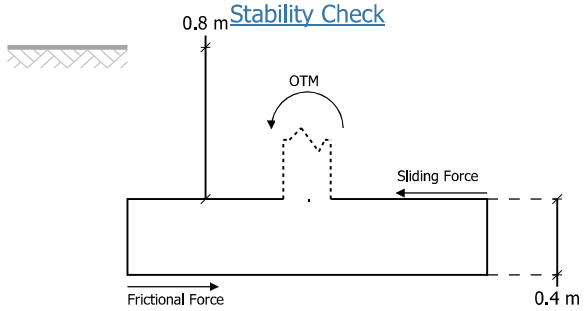
Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
46	**100.5419**	40.2323	0.3488	60.6585	0.000
22	16.3087	**81.5970**	80.6316	15.3434	0.000
22	16.3087	81.5970	**80.6316**	15.3434	0.000
23	55.9476	7.3737	40.5745	**89.1483**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
46	**100.5419**	40.2323	0.3488	60.6585
22	16.3087	**81.5970**	80.6316	15.3434
22	16.3087	81.5970	**80.6316**	15.3434
23	55.9476	7.3737	40.5745	**89.1483**

Stability Check



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
10	240.867	42.151	58.718	1.500	89.335	208.523	1.500
11	240.867	42.151	58.718	1.500	89.335	208.523	1.500
12	293.902	41.117	57.587	1.500	87.155	213.000	1.500
13	278.562	41.371	57.872	1.500	87.690	211.863	1.500
14	234.172	36.850	51.480	1.500	19.025	203.059	1.500
15	31.321	42.144	35.552	1.500	88.500	23.427	1.500
16	6.076	59.861	8.549	1.500	181.158	6.046	1.500
17	7.380	15.498	9.423	1.500	12.112	7.972	1.500
18	3.541	192.419	5.006	1.500	642.303	3.513	1.500
19	4.393	7.512	5.363	1.500	6.600	4.794	1.500
20	23.549	55.559	30.663	1.500	115.962	17.768	1.500
21	303.787	21.529	30.370	1.500	13.839	264.225	1.500
22	4.528	72.838	6.392	1.500	301.253	4.489	1.500
23	5.530	10.518	6.923	1.500	8.722	6.008	1.500
39	8.348	59.846	11.692	1.500	197.205	8.428	1.500
40	6.490	8.958	7.433	1.500	9.763	6.648	1.500
41	4.991	191.499	7.057	1.500	433.260	5.088	1.500
42	3.944	6.109	4.686	1.500	6.249	4.071	1.500
45	6.267	72.789	8.830	1.500	367.450	6.363	1.500
46	4.915	7.164	5.732	1.500	7.589	5.056	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 18

Governing Disturbing Force : 20.635kN

Governing Restoring Force : 73.061kN

Minimum Sliding Ratio for the Critical Load Case : 3.541

Critical Load Case for Overturning about X-Direction :	42
Governing Overturning Moment :	-22.041kNm
Governing Resisting Moment :	137.730kNm
Minimum Overturning Ratio for the Critical Load Case :	6.249

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction :	42
Governing Disturbing Force :	-12.578kN
Governing Restoring Force :	76.840kN
Minimum Sliding Ratio for the Critical Load Case :	6.109
Critical Load Case for Overturning about Z-Direction :	18
Governing Overturning Moment :	-37.643kNm
Governing Resisting Moment :	132.238kNm
Minimum Overturning Ratio for the Critical Load Case :	3.513

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

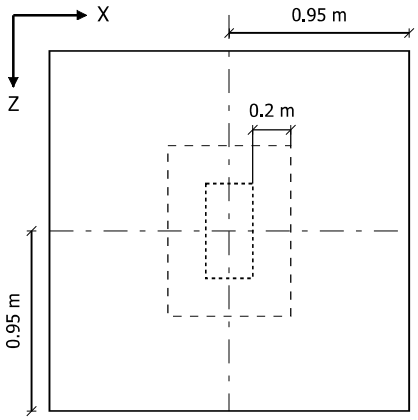
Critical Load Case for Sliding along Resultant Direction :	42
Governing Disturbing Force :	23.189kN
Governing Restoring Force :	108.668kN
Minimum Sliding Ratio for the Critical Load Case :	4.686

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.400m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.344m

For rectangular column, β_c = B_{col} / D_{col} = 2.000

Effective depth, d_{eff}, increased until 0.75XV_c $\geq$ Punching Shear Force

Punching Shear Force, Vu = 193.634kN, Load Case # 50

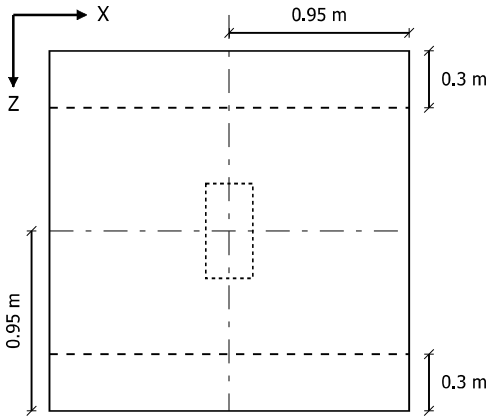
From ACI Cl.11.11.2, b_o for column=	$2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ =	2.876m
Equation 11-31, V_{c1} =	$0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$	1944.021kN
Equation 11-32, V_{c2} =	$0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$	3297.265kN
Equation 11-33, V_{c3} =	$0.33 \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$	1944.021kN
Punching shear strength, V_c =	0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) =	1458.016kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 642.149kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{zd}$ = 0.356m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

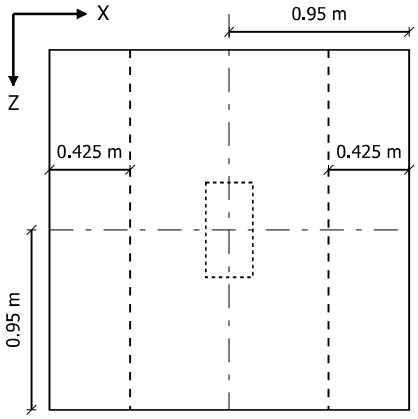
From above calculations, 0.75 X V_c = 481.612 kN

Critical load case for V_{ux} is # 50 $V_{ux} = V_{ux}|_{x=D_x}$ = 55.275 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 642.149 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.481 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

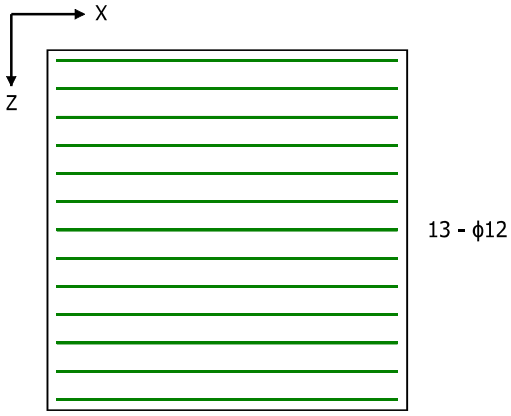
From above calculations, 0.75 X V_c = 481.612 kN

Critical load case for V_{uz} is # 29 $V_{uz} = V_{uz}|_{z=D_z} =$ 84.425 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 29

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.344 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488
From ACI Cl. 7.12.2, ρ_{min} =		0.00166
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	1.075 m
Ultimate moment,	$M_u\|_{z=D_z}$ =	58.515 kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	65.016 kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00069
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00060
Since	ρ= ρ_{min}	ρ_{min} Governs
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	1264.425 mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 125.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars =	$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ =	=0.410	m
Available development length for bars, D_L =	$0.5 \times (L - D_{col}) \cdot - C_{cover}$ =	0.775	m
Try bar size # 12	Area of one bar =	113.097	mm2
Number of bars required, N_{bar} =	$\frac{A_s}{A_{bar}}$ =	13	

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} =	N_{bar} X (Area of one bar) =	1470.255	mm2
d_{eff} =	D - C_{cover} - 0.5 X (dia. of one bar) =	0.344	m
Reinforcement ratio, ρ =	$\frac{A_{s_total}}{(d_{eff} \times W)}$ =	0.00225	

From ACI Cl.7.6.1, minimum req'd clear distance between bars

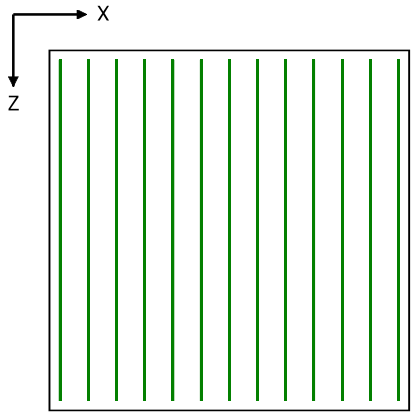
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.163

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



13 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 50

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.332 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488

From ACI Cl.7.12.2, ρ_{min} = 0.00162

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{\left(0.85 \times F_c'\right)}$ = 14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 0.700 m

Ultimate moment, $M_u\big|_{x=D_x}$ = 37.189 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 41.322 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00047

(Based on gross depth) ρ x d_{eff} / Depth = 0.00039

Since ρ= ρ_{min} ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 1233.789 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 125.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times \left(L - D_{col}\right) - C_{cover}$ = 0.650 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 13

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1470.255 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.332 m

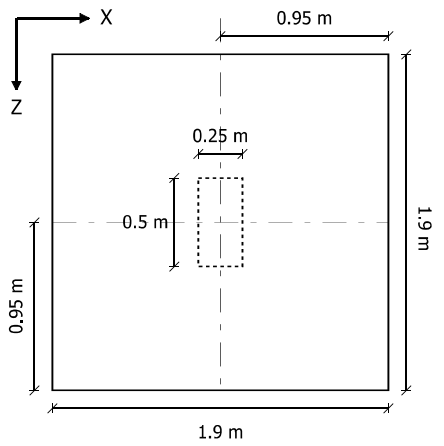
Reinforcement ratio, ρ = $\frac{A_{s_total}}{\left(d_{eff} \times W\right)}$ = 0.00233

From ACI Cl.7.6.1, minimum req'd clear distance between bars

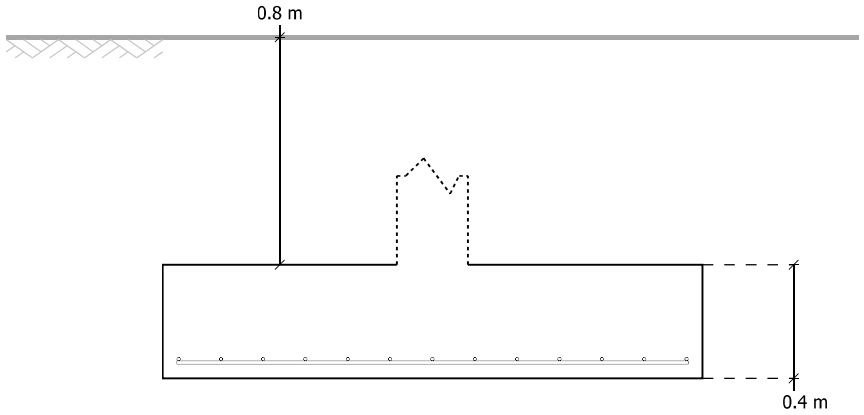
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.192

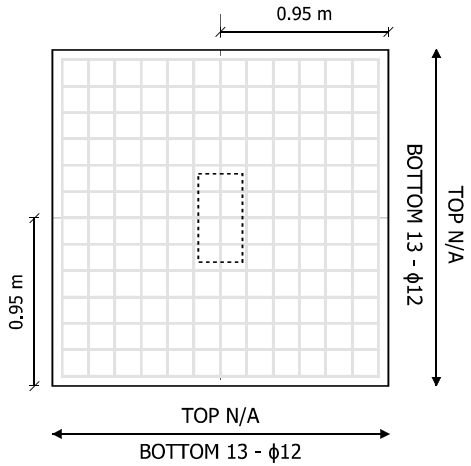
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

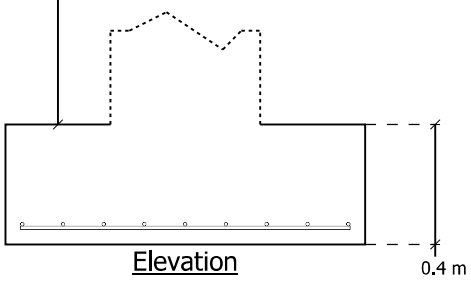
	Dimensions (m)			Reinforcement			
	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
Foundation	1.9 m	1.9 m	0.4 m	N/A	N/A	13 - ϕ12	13 - ϕ12
	l	b	h	Main Reinforcement		Trans Reinforcement	
Pedestal	N/A	N/A	N/A	N/A		N/A	

Print Calculation Sheet

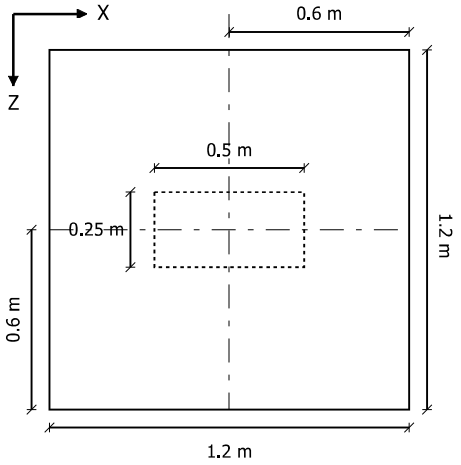
FOUNDATION-F1 ALONG GRID 4 & A

Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
47	1	1.200m	1.200m	0.400m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
47	9 - ϕ12	9 - ϕ12	N/A	N/A	N/A	N/A



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 400.000mm
Footing Length - X (Fl) : 1200.000mm
Footing Width - Z (Fw) : 1200.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.500m

Column Width - Z (B_{col}) : 0.250m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 125.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 800.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.530
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 1.200m

Initial Width (W_o) = 1.200m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
10	SLS: (DL)	1.00	1.00	1.00
11	SLS: (DL+T)	1.00	1.00	1.00
12	SLS: (D+LR)	1.00	1.00	1.00
13	SLS: (D+0.75LR)	1.00	1.00	1.00
14	SLS: (D+0.7EZ)	1.00	1.00	1.00
15	SLS: (D+0.7EX)	1.00	1.00	1.00
16	SLS: (D+0.75WX1+0.75LR)	1.00	1.00	1.00
17	SLS: (D+0.75WZ1+0.75LR)	1.00	1.00	1.00
18	SLS: (0.6D+WX1)	1.00	1.00	1.00
19	SLS: (0.6D+WZ1)	1.00	1.00	1.00
20	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
21	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
22	SLS: (D+WX1)	1.00	1.00	1.00
23	SLS: (D+WZ1)	1.00	1.00	1.00
39	SLS: (D+0.75WX2+0.75LR)	1.00	1.00	1.00
40	SLS: (D+0.75WZ2+0.75LR)	1.00	1.00	1.00
41	SLS: (0.6D+WX2)	1.00	1.00	1.00
42	SLS: (0.6D+WZ2)	1.00	1.00	1.00
45	SLS: (D+WX2)	1.00	1.00	1.00
46	SLS: (D+WZ2)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
24	ULS: (1.4 DL)	1.00	1.00	1.00
25	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
26	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+0.80WX1+1.6LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+1.6WX1+0.5LR)	1.00	1.00	1.00
30	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
31	ULS: (1.2D+EX)	1.00	1.00	1.00
32	ULS: (1.2D+EZ)	1.00	1.00	1.00
33	ULS: (0.90D+1.6WX1)	1.00	1.00	1.00
34	ULS: (0.90D+1.6WZ1)	1.00	1.00	1.00
35	ULS: (0.90D+EZ)	1.00	1.00	1.00
36	ULS: (0.90D+EX)	1.00	1.00	1.00
47	ULS: (1.2D+0.80WX2+1.6LR)	1.00	1.00	1.00
48	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
49	ULS: (1.2D+1.6WX2+0.5LR)	1.00	1.00	1.00
50	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
53	ULS: (0.90D+1.6WX2)	1.00	1.00	1.00
54	ULS: (0.90D+1.6WZ2)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
10	167.467	-5.856	15.567	5.562	5.256
11	167.467	-5.856	15.567	5.562	5.256
12	170.465	-5.500	15.611	5.573	5.293
13	169.716	-5.589	15.600	5.570	5.284
14	171.253	-5.854	18.039	8.327	5.257
15	165.525	-3.090	15.504	5.533	0.065
16	168.551	-1.426	13.192	4.905	-3.255
17	175.760	-5.066	17.624	8.838	5.771
18	98.927	2.038	6.130	2.450	-8.231
19	108.540	-2.816	12.040	7.694	3.803

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
20	98.539	-0.748	9.277	3.308	-2.038
21	104.266	-3.512	11.813	6.103	3.155
22	165.914	-0.304	12.356	4.675	-6.128
23	175.526	-5.158	18.266	9.918	5.905
39	174.729	-10.466	13.477	5.036	13.986
40	163.713	-4.484	14.586	2.687	5.521
41	107.165	-10.016	6.510	2.625	14.756
42	92.476	-2.039	7.989	-0.507	3.470
45	174.152	-12.358	12.737	4.850	16.859
46	159.463	-4.381	14.215	1.718	5.572

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
24	234.454	-8.198	21.793	7.787	7.358
25	235.953	-8.020	21.815	7.792	7.376
26	235.953	-8.020	21.815	7.792	7.376
27	204.515	-2.017	16.182	5.982	-2.741
28	215.355	-5.733	23.021	13.650	7.365
29	199.974	2.033	13.566	5.260	-11.889
30	215.355	-5.733	23.021	13.650	7.365
31	198.187	-3.076	18.590	6.633	-1.109
32	206.369	-7.024	22.212	10.625	6.309
33	148.235	3.612	8.874	3.586	-13.485
34	163.615	-4.154	18.329	11.976	5.769
35	156.129	-5.268	17.542	8.956	4.732
36	147.947	-1.319	13.920	4.964	-2.686
47	211.105	-11.660	16.487	6.123	15.649
48	189.653	-4.490	16.540	0.529	6.832
49	213.155	-17.253	14.175	5.541	24.890
50	189.653	-4.490	16.540	0.529	6.832
53	161.416	-15.673	9.483	3.867	23.295
54	137.914	-2.911	11.848	-1.145	5.236

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 1.440 m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 0.984 m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 1.200 m Governing Load Case : # 10

Width (W_2) = 1.200 m Governing Load Case : # 10

Depth (D_2) = 0.400 m Governing Load Case : # 49

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 1.440 m^2

Final Soil Height = 0.800 m

Footing Self Weight = 13.824 kN

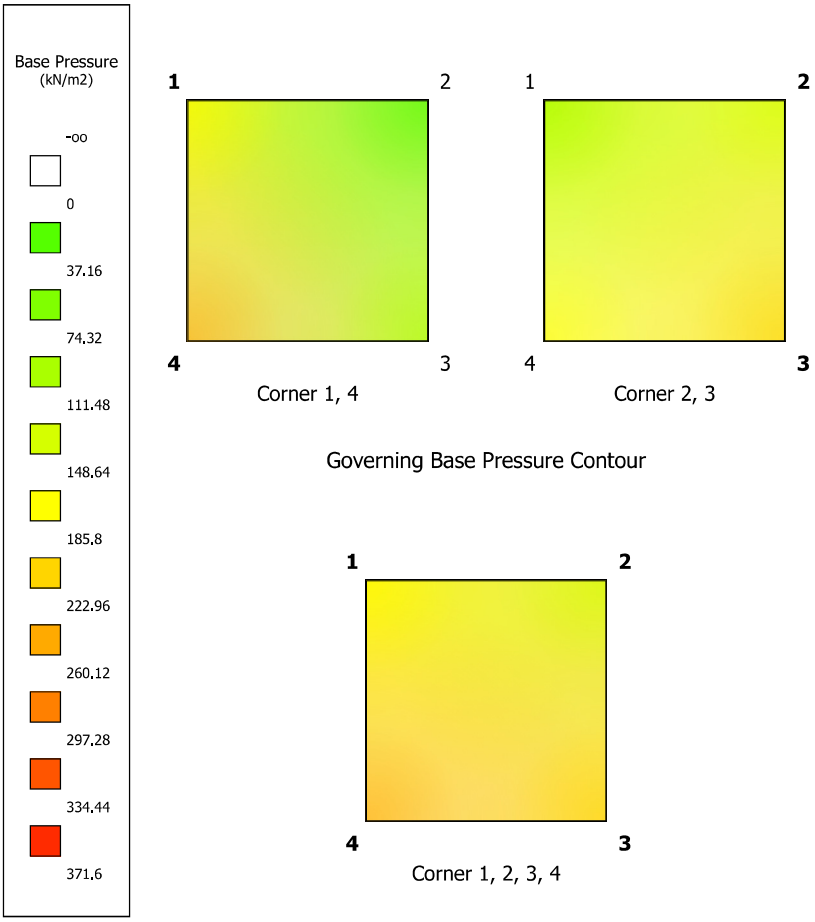
Gross Soil Bearing Capacity = 371.60 kN/m2

Soil Weight On Top Of Footing = 18.935 kN

Pressures at 4 Corners

Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner



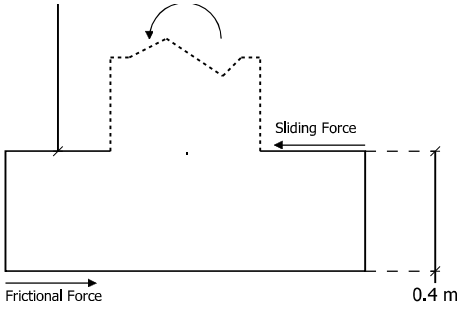
Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
45	**184.8578**	33.4550	102.5182	253.9209	0.000
22	83.7160	**125.4305**	192.2182	150.5037	0.000
22	83.7160	125.4305	**192.2182**	150.5037	0.000
45	184.8578	33.4550	102.5182	**253.9209**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
45	**184.8578**	33.4550	102.5182	253.9209
22	83.7160	**125.4305**	192.2182	150.5037
22	83.7160	125.4305	**192.2182**	150.5037
45	184.8578	33.4550	102.5182	**253.9209**

Stability Check



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
10	18.123	6.817	9.024	1.500	10.259	15.811	1.500
11	18.123	6.817	9.024	1.500	10.259	15.811	1.500
12	19.582	6.900	9.203	1.500	10.387	16.273	1.500
13	19.200	6.879	9.158	1.500	10.355	16.156	1.500
14	18.471	5.994	8.063	1.500	7.927	16.109	1.500
15	34.011	6.778	9.401	1.500	10.208	91.481	1.500
16	74.840	8.088	11.372	1.500	11.943	44.996	1.500
17	21.814	6.271	8.523	1.500	7.926	16.046	1.500
18	34.247	11.386	15.280	1.500	16.284	8.734	1.500
19	26.593	6.220	8.565	1.500	6.842	17.199	1.500
20	93.073	7.501	10.574	1.500	11.339	45.309	1.500
21	20.681	6.148	8.334	1.500	7.668	18.032	1.500
22	346.029	8.522	12.048	1.500	12.479	19.844	1.500
23	21.401	6.043	8.225	1.500	7.302	15.683	1.500
39	10.507	8.159	9.114	1.500	12.017	6.851	1.500
40	23.225	7.139	9.650	1.500	13.928	16.117	1.500
41	7.404	11.391	8.780	1.500	16.209	4.475	1.500
42	32.549	8.309	11.385	1.500	28.249	17.533	1.500
45	8.874	8.610	8.739	1.500	12.565	5.694	1.500
46	23.252	7.167	9.686	1.500	15.686	15.746	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 41
Governing Disturbing Force : -10.016kN
Governing Restoring Force : 74.160kN
Minimum Sliding Ratio for the Critical Load Case : 7.404
Critical Load Case for Overturning about X-Direction : 19
Governing Overturning Moment : 12.509kNm
Governing Resisting Moment : 85.587kNm
Minimum Overturning Ratio for the Critical Load Case : 6.842

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 14
Governing Disturbing Force : 18.039kN
Governing Restoring Force : 108.126kN

Minimum Sliding Ratio for the Critical Load Case : 5.994

Critical Load Case for Overturning about Z-Direction : 41

Governing Overturning Moment : 18.762kNm

Governing Resisting Moment : 83.953kNm

Minimum Overturning Ratio for the Critical Load Case : 4.475

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

Critical Load Case for Sliding along Resultant Direction : 14

Governing Disturbing Force : 18.965kN

Governing Restoring Force : 152.914kN

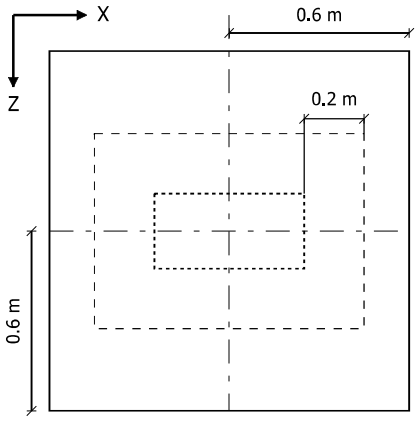
Minimum Sliding Ratio for the Critical Load Case : 8.063

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.400m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.344m

For rectangular column, β_c = B_{col} / D_{col} = 2.000

Effective depth, d_{eff}, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 175.160kN, Load Case # 25

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 2.876m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 1944.021kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 3297.265kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 1944.021kN

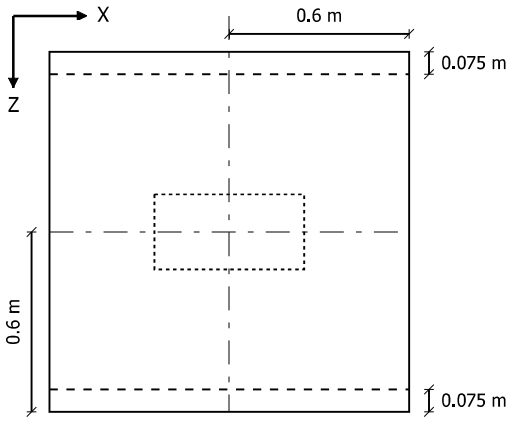
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1458.016kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 405.568kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{xd}$ = 0.131m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

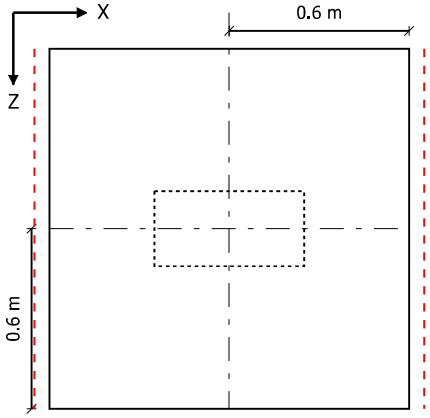
From above calculations, 0.75 X V_c = 304.176 kN

Critical load case for V_{ux} is # 30 $V_{ux} = V_{ux}|_{x=D_x}$ = 38.201 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 405.568 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.006 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

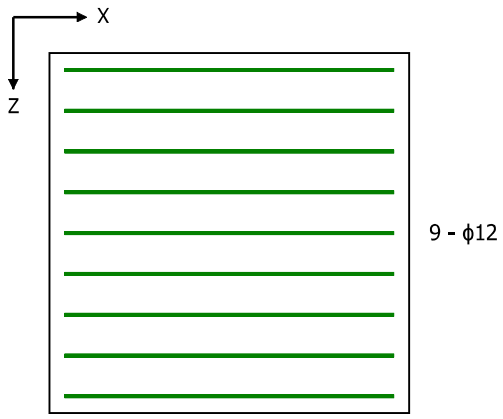
From above calculations, 0.75 X V_c = 304.176 kN

Critical load case for V_{uz} is # 49 $V_{uz} = V_{uz}|_{z=D_z}$ = 2.020 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 49

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.344	m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796	
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317	
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488	
From ACI Cl. 7.12.2, ρ_{min} =		0.00174	
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118	

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	0.350	m
Ultimate moment,	$M_u\big\vert_{z=D_z}$ =	19.086	kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	21.207	kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00036	
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00031	
Since	ρ= ρ_{min}	ρ_{min} Governs	
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	833.896	mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 125.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = =0.410 m

$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}} =$

Available development length for bars, D_L = $0.5 \times (L - D_{col}) - C_{cover} =$ 0.300 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, $N_{bar} = \frac{A_s}{A_{bar}} =$ 9

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1017.869 mm2

d_{eff} = D - C_{cover} - 0.5 X (dia. of one bar) = 0.344 m

Reinforcement ratio, $\rho = \frac{A_{s_total}}{(d_{eff} \times W)} =$ 0.00247

From ACI Cl.7.6.1, minimum req'd clear distance between bars

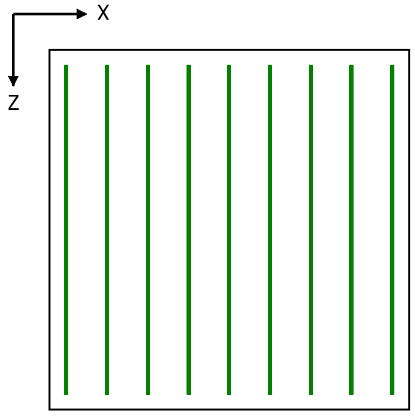
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.221

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



9 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 30

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} = 0.332 m

Factor β_1 from ACI Cl.10.2.7.3 = 0.796

From ACI Cl. 10.3.2, $\rho_{bal} = 0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$ 0.03317

From ACI Cl. 10.3.3, $\rho_{max} = 0.75 \times \rho_{bal} =$ 0.02488

From ACI Cl.7.12.2, ρ_{min} = 0.00177

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{(0.85 \times F_c')} =$ 14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 0.725 m

Ultimate moment, $M_u|_{x=D_x}$ = 31.242 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 34.714 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00063

(Based on gross depth) ρ x d_{eff} / Depth = 0.00052

Since $\rho = \rho_{min}$ ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 848.761 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 125.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times (L - D_{col}) - C_{cover}$ = 0.425 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 9

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1017.869 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.332 m

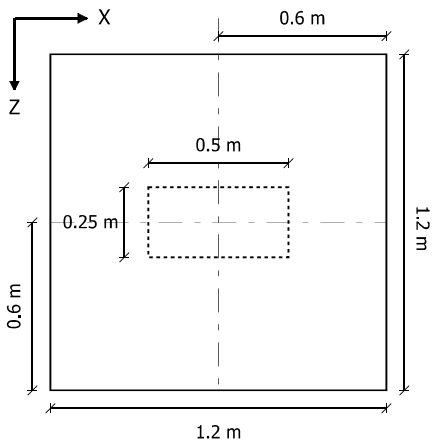
Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00255

From ACI Cl.7.6.1, minimum req'd clear distance between bars

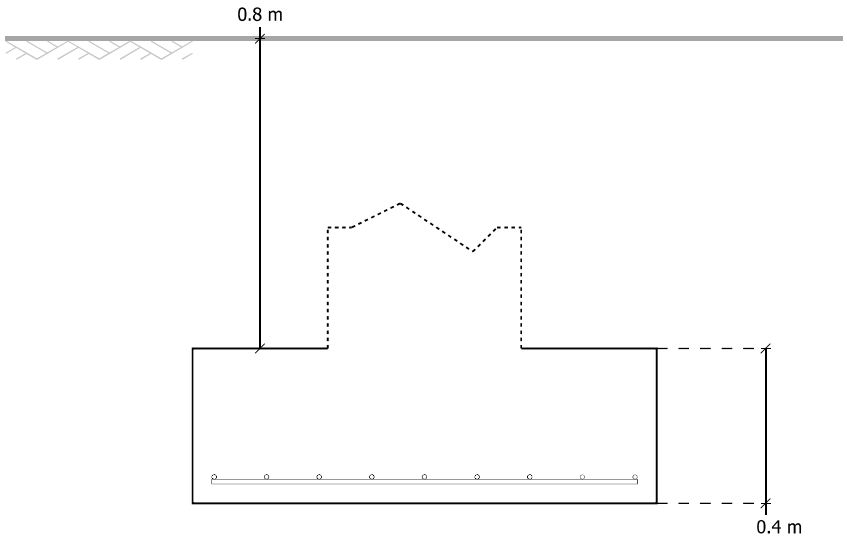
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.199

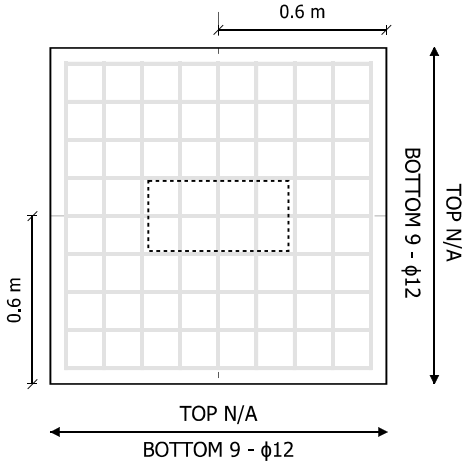
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

	Dimensions (m)			Reinforcement			
	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
Foundation	1.2 m	1.2 m	0.4 m	N/A	N/A	9 - ϕ12	9 - ϕ12
	l	b	h	Main Reinforcement		Trans Reinforcement	
Pedestal	N/A	N/A	N/A	N/A		N/A	

Print Calculation Sheet